Department for
Science, Innovation
& Technology

Public Emergency Call Service

Code of Practice 'OFFICIAL' Version

Version 1.1

Acknowledgements

Thank you to the members of the 999/112 Liaison Committee for providing the updates and content for this document.

Any enquiries regarding this publication should be sent to us at telecoms.resilience@dsit.gov.uk

Contents

1. General Information 6
1.1 Document Overview 6
1.2 Document Ownership and Review 6
1.3 Requirements of the Communication Provider 6
2. Quality of Service 8
2.1. Call Handling Agent Standards 8
2.2. Emergency Authority Standards 8
3. Process for Caller Connection to the Correct EA 9
3.1 The Calling Customer 9
3.2 Multi-Agency Calls 9
3.3 EA Selection 9
3.4 Connection To and From International EA Control Rooms 10
3.5 Delivery of the Call to EA Control Rooms 11
4. Difficulties Connecting to the EACR 13
4.1 Critical Contact Number 13
4.2 Buddy EA 13
5. Contingencies and Review of EACR Numbers 14
6. Call Queuing Systems at EACRs 15
6.1 Queue Set Up 15
6.2 Queue Announcement/Message 15
7. Call Monitoring and Disconnection 17
7.1 EA Requires Further Information 18
8. Calls Without Service Request 19
8.1 Calls Without Service Request from Mobile Phones 19
8.2 112 Calls Without Service Request from Fixed Phones 19
8.3 999 Call Connection Flow Chart 21
9. Post-call Provision of Call and Caller Information 22
9.1 Processes 22
9.2 The Emergency Period 22
9.3 Confidential Customer Information 23

9.4 Mobile Phone Subscriber Checks 23
9.5 Mobile Virtual Network Operator (MVNO) 24
10. Misrouted Calls and Mobile Zone Problems 25
10.1 Boundary Overlap (Mobile ECNs) 26
10.2 Failure to Display Zone Or Cell Id Information (Mobile ECNs) 26
11. Routing Methods and Fault Management 28
11.1 Routing Methods and Network Security 28
11.2 Priority Fault Repair Service 29
11.3 Isolation of Customers Due to Network Failure 29
11.4 Draft Media Statement for CHAs/Cps and EAs 29
12. Management Of Major Influxes of Emergency Calls 30
12.1 Advance Warning of Major Incidents 30
12.2 Contacts 30
12.3 Modifications to Queueing Systems 30
12.4 Changes to Call Handling Procedures 31
12.5 Temporary Procedure Contingency 31
12.6 OPERATION WILLOW BECK 32
12.7 Flooding and Major Incidents 34
12.8 Media Announcements During a Major Incident 34
13. Recording of Calls 35
14. Guidance on Public Advice 36
14.1 Broad Principles of Messaging 36
14.2 Channels of Communications 36
14.3 Suggested Lines to the Public During Outage Scenarios 36
15. Other Types Of Calls 43
15.1 Limited Service State Emergency Calls 43
15.2 RELAY UK (Previously Next Generation Text (NGT)/Text Relay/Typetalk) 44
15.3 999 British Sign Language Emergency Video Relay Service 45
15.3.1 999 BSL App and Web Access 45
15.4 EmergencySMS (eSMS) 46
15.5 Third Party Call Centre Emergency Calls 46
15.6 eCall And Telematics Emergency Calls 47
15.7 Other Vehicle Activations 48

15.8 Google 'Loss of Pulse' 48
APPENDIX 49
APPENDIX A: GLOSSARY OF TERMS 50
APPENDIX B: VERSION CONTROL 52
APPENDIX C: GUIDANCE ON USE OF LOCATION INFORMATION 53
APPENDIX D: BEST PRACTICE GUIDE FOR CPs ON NAME & ADDRESS INFORMATION (LANDLINES) 57
APPENDIX E: MOBILE DISCONNECTION 59
APPENDIX F: COMMUNICATIONS DATA CODE OF PRACTICE (PURSUANT TO IPA) 60
APPENDIX G: eSMS MEMORANDUM OF UNDERSTANDING FOR PEOPLE WHO ARE DEAF, HARD OF HEARING or WHO HAVE IMPAIRED SPEECH 62

1. General Information

1.1 Document Overview

This Code of Practice ("the Code") is intended to advise organisations on the method of handling '999/112' public emergency telephone calls between the Call Handling Agents ("CHAs") and the Emergency Authorities ("EAs") in the UK (Police, Fire, Ambulance and Coastguard). It does not reflect HMG policy.

This document is the OFFICIAL version for those Communication Providers ("CPs") who do not sit on the Liaison Committee but require information on their responsibilities to the CHAs and EAs. This document is not for wider distribution unless agreed with the document owner. Sensitive information has been removed including; personal information protected by the Data Protection Act, commercial interests, specifics on law enforcement and certain health and safety information.

The Call Handling Agents and the Emergency Authorities have agreed the contents of the Code through the auspices of the 999/112 Liaison Committee. The Code is subject to regular review by the 999 Liaison Committee.

1.2 Document Ownership and Review

This document is owned by the Department for Science Innovation and Technology on behalf of the 999/112 Liaison Committee. It will be reviewed at least annually.

1.3 Requirements of the Communication Provider

Under the regulatory regime set out in the Communications Act 2003, communications providers do not require a licence to operate in the UK, but can be made subject to conditions of general application. All Communications Providers have obligations under the Communications Act 2003, implemented under Condition A3 of Ofcom's General Conditions of Entitlement ("GC's") in respect of emergency calls made to 999 and 112. Communications Providers must ensure they comply with all aspects of these obligations. The most up to date version can be accessed at https://www.ofcom.org.uk/phones-and-broadband/accessibility/general-conditions-of-entitlement

The CP must ensure that any end user can access EAs by using the emergency numbers "112" and "999" at no charge and, in the case of a pay telephone, without having to use coins or cards. From 2009 Ofcom expected all UK mobile network operators to introduce the ability for users to make calls to 999 or 112 on another mobile network – thereby allowing users to make emergency calls when out of the coverage of their home network to maximise the ability for users to obtain help quickly. These calls are variously known as 'displaced' calls, "emergency roamers", "camped" calls, or Limited Service State calls. These calls will be referred to as Limited Service State Calls throughout this document. CHA operators will refer to them as Limited Service State calls at hand over to the EA control rooms to quickly convey that the call is not on its home network. This functionality should not be confused with foreign mobiles visiting the UK as roaming agreements would be in force that would allow the user to make and receive calls as normal.

As per Ofcom's General Condition’s GC A3.5 and A3.6, the CP must also make caller location information available to the EAs, to the extent technically feasible, for all calls to the emergency numbers 112 and 999. ‘Caller Location’ is defined as any data or information processed in an Electronic Communications Network (“ECN”) which gives the geographic position of the terminal equipment of the person making the emergency call. Note in the case of Limited Service State calls no valid Calling Line Identity (CLI) will be passed from the ECN.

The CPs deal with the requirement to provide a Public Emergency Call Service by contracting with a Call Handling Agent (CHA). In the UK BT currently acts as CHA for all the networks.

The handling of an emergency call by a CHA involves the five main phases:

- Connection of the caller over the CP and CHA networks to the CHA’s Emergency Operator (“EO”) via the 999/112 number.
- Selection by the emergency operator of the required EA Control Room (“EACR”).
- Onward connection of the caller to the EACR over the CP/CHA networks.
- Listening by the EO to ensure that connection has been established with the appropriate EACR and the ability to provide further assistance to the caller or Emergency Authority (EA) when required.
- Provision of location information

2. Quality of Service

2.1. Call Handling Agent Standards

Under present legislation there are no quality-of-service standards laid down for 999/112. However, Ofcom does set a Key Performance Indicator for the answering of 999/112 calls to ensure a consistently good service for users call handling by CHAs or the EAs. The CHAs have a target of answering 95% of 999/112 calls within 5 seconds (“PCA5”).

To meet this commitment, 999/112 calls receive priority by CHA EOs over all other calls. If the EO has any other call in hand when a 999/112 call arrives the operator will ask that caller to hold the line and will deal with the emergency call. In the same manner, the EAs place the highest priority on incoming emergency code calls. All such calls are treated equally with no discrimination between fixed line and mobile callers.

2.2. Emergency Authority Standards

EAs are subject to recommended call answering response times.

The Police service in England, Wales and Northern Ireland aim to answer 90% of 999/112 calls within 10 seconds, and Police Scotland aim to have a mean answering time of under 10 seconds for all 999/112 calls;

The recommended response time for the Ambulance Service is to answer 95% within 5 seconds, Scottish Ambulance 90% in 10 seconds;

The Fire and Rescue Service and the Coastguard aim to answer 95% within 10 seconds.

CHAs and EAs performance will be presented (through figures gathered centrally) at each 999/112 Liaison Committee meeting.

3. Process for Caller Connection to the Correct EA

3.1 The Calling Customer

Before connecting a call to the relevant EA, CHA EOs will obtain, or seek to obtain, from the 999/112 caller

which EA the caller requires (i.e. fire, police, ambulance, coastguard).

the telephone number of the caller if the caller's telephone number is not automatically available. If the caller cannot provide it, the EO will ask for the approximate location of the caller and seek to connect the call to the appropriate EA. It is the responsibility of the EACR staff to obtain adequate address information from the caller to enable the EACR to locate the incident being reported.

3.2 Multi-Agency Calls

If a customer makes a request for more than one EA the CHA operator will connect the customer to the first EA requested and provide a verbal handover stating that the customer has requested more than one service (and advise which additional services are required).

The EA controller will confirm to the CHA operator during call handling that they will pass details to the other EA(s) and do so using existing processes. If the EA controller is unable to do this on occasions, they will request the CHA operator to remain on the call and onward connect to the other requested EA(s). They may ask the caller to stay online at the end of the call so that, when the EA clears, the call will be represented to the CHA 999 operator for further connection to the other service(s). When the first EA ends their call, and the caller stays online, the call will be represented back to the CHA, who will then ask which service is needed and connect to the next EA required. However, this service will only work if the call was connected to the EA over a non-analogue connection and on the CHA 999 primary platform.

3.3 EA Selection

When a CHA EO answers an emergency call, the full national calling number (except in the case of Limited Service State calls – see Section 15.1) and for mobile calls the zone code (group of cells) or Cell ID, will normally be automatically displayed on the operator's console. For 999/112 calls this calling number will be displayed on the operator's console even if the caller withheld their number.

The EACR routings automatically displayed are based on matching the EACR areas to the calling number's postal code (or in absence of a postal code the exchange catchment area or nearest town information) for fixed calls, and to the zone code or Cell ID for mobile calls. The list of EACR routings shown is agreed between the CPs, the CHAs and the EAs. This information is updated as necessary to account for number changes, new cells etc.

There are a small number of cases where callers will need to be asked for their number, or asked to confirm their location, in order to display the correct EACR routing. These cases include (a) Alarm Monitoring Centres forwarding calls from their clients, (b) when a Voice over

Internet Protocol (VoIP) terminal is being used, (c) when a default number has displayed due to an exchange working in fall-back mode (d) where a PBX extension user is displaying the main number.

3.4 Connection To and From International EA Control Rooms

Some calls may require contact with EA control rooms in other countries, and conversely international EA controls rooms may need to contact counterparts in the UK. A confidential directory of international contact numbers has been created and managed by the European Communications Office (ECO*).

3.4.1 Calls from Internation EA Control Room into the UK

The CHA will provide a number to the ECO directory to allow calls from international EAs to route to the CHA. These calls will be handled by the CHA emergency operator in the same way as any other incoming emergency call and will ask for both the emergency service required and the most appropriate city/town in which assistance is required. If no service and/or location information is given, then the CHA EO will follow existing procedures for UK calls for which no service and/or location information is provided. On connection with the EA control room, the CHA EO will announce that the call has been received from an international EA control room.

3.4.2 Calls to UK EA Control Rooms Requiring Assistance Abroad

The CHA will forward all UK emergency calls to the appropriate UK EA control room as described in section 3.3 above. It is for the EA Operator to assess any call and determine whether assistance from an international EA is required.

If an EA operator determines that assistance is required from emergency authorities in another country, the EA will make a separate 999 call and request connection with an EA in the appropriate country. The BT CHA EO will use the ECO directory to look up the relevant number and establish contact. The CHA EO will stay on the line until two-way conversation between the two control rooms has begun (in line with conventional UK calls).

3.4.3 Calls for Countries not in the ECO List

It is the responsibility of the EA to connect calls to countries not listed, the EA must either provide the necessary contact numbers or arrange for communication through alternative methods.

3.4.4 Roles and Responsibilities

The CHA will:

- Ensure that the latest version of the ECO directory is available to EO's;
- Ensure that UK details (names and contact numbers) are correctly maintained in the directory; and
- Use the directory to make contact with international EAs when instructed to do so by UK EA control room operators.

The EA operators will:

- Assess emergency calls and determine whether assistance abroad is required and in which country;
- Contact the CHA via a separate call to 999 to request connection to international EAs; and
- Direct the CHA to manage exceptions in this process – for example if calls cannot be established because there is no contact number for the country required or if the number given is unobtainable. The responsibility to connect calls to countries not listed remains with the EA to either provide contact numbers or for them to contact via a different means.

* Background information on the functions of the European Communications Office can be found here: https://www.cept.org/eco/eco-in-brief. Please note that it is not an EU institution.

3.5 Delivery of the Call to EA Control Rooms

It is the responsibility of the EAs to provide the means of receiving emergency calls and keep the 999/112 liaison contact point in the CHA informed of the equipment and connect-to routings in use in the EACRs. To cater for unforeseen circumstances EAs will provide three separate routes.

3.5.1 Primary Route

This is the route that the CHA operator will initially use to connect a caller to the EACR and the EA must provide sufficient capacity on this route to handle normal 999/112 traffic distribution. EAs will reserve primary routes exclusively for receiving 999/112 calls.

3.5.2 Secondary

Applicable for EA's with the exception of Ambulance Trusts (see 3.5.4 for more information). In circumstances where the CHA emergency operator receives no reply on the primary number after 60 seconds (or 2 minutes where a queue is in use) the operator will connect the call to a secondary number provided by the EA. This procedure should only be necessary in instances when the EACR has an unusually high level of traffic, or a fault in its system or in one of the ECNs.

3.5.2 Alternative

Applicable for EA's with the exception of Ambulance Trusts (see 3.5.4 for more information). In the event of a major problem which results in the primary and secondary routes to an EACR being unavailable to the CHAs, including a further period of 30 seconds of no reply on the secondary number, the EA should provide the CHAs with an alternative means of taking delivery of the call.

To provide adequate resilience this alternative number must be served by a different network route from that providing the primary and secondary routes, ideally at a different EACR for maximum resilience. EAs must consider, where appropriate, which EACRs are used as alternatives to each other.

3.5.4 Ambulance Trust Delivery

All calls connected to Ambulance Trusts are connected through an Intelligent Routing

Platform (“IRP”). If the IRP calculates that the “home” ambulance trust is likely to be unable to answer the call within an agreed threshold (currently 180 seconds), the IRP will re-route to an alternative ambulance Trust if there is availability. As further back up in the unlikely event the call is not re-routed via the IRP then the CHA will connect the call to the IRP re-routing number. The IRP has business rules built into the platform using data fed directly from the ambulance trust control telephony systems and will reroute calls as described above. If transmission problems are encountered, the CHA will reconnect to the next number. If the call is identified as critical, the call will be connected to the secondary line.

Notes

- Mobile CPs and the 999/112 Liaison Committee do not advise the use of a mobile phone as the ‘alternative’.
- The EA and its CP(s) need to liaise regularly to ensure that there is no single point of failure affecting both the primary and secondary routes.
- The same primary, secondary and alternative numbers should be used for all 999/112 calls, whether fixed or mobile, and regardless of which CHA connects the call.
- All the routes will need to be staffed on a 24-hour basis.

4. Difficulties Connecting to the EACR

4.1 Critical Contact Number

EAs need to provide the CHAs 999/112 liaison contact point with a 24hr critical contact number (CCN) within the EACR to be used only when Operator Centres are having difficulty connecting calls to the EACR. This number will only be used for CHA Operator Centre managers to contact EACR managers in the event of 999/112 call handling problems relating to call surges, call answering times, staffing difficulties, or other problems so that corrective action can be agreed. For example, Operator Centre managers will call if there are extended delays in an EACR answering of about 5 minutes or where there are delays and they have an exceptionally critical call waiting.

Note

To ensure EACR managers receive the minimum number of calls, it will normally only be one BT centre that calls, and they should then cascade relevant information to other centres.

If the CHA Manager is unable to contact the EACR Manager on their Critical Contact number, then they will speak to: -

- a neighbouring EACR of the same type (preferably a "buddy EA" – see below) as first choice to see if they can help by either offering callers advice on line, or passing prioritised details back to the original EA (perhaps using radio), or maybe even responding in border areas; or
- an EACR that covers the same area (but of a different type) as second choice as they may be able to pass prioritised details back to the original EA (perhaps using radio), or will at least be able to offer more help than the CHA to very distressed callers.

4.2 Buddy EA

EAs need to pre-nominate an appropriate "Buddy EA" for the CHAs to use. "Buddy EAs" are expected to be used to assist in times of unexpected pressure and improve the speed of implementation and the effectiveness of contingency arrangements at such times by, for example, selecting "buddies" which allow call data to be readily transferred back to the original EACR. Ambulance Trusts use of other/buddy trusts is through IRP as detailed in section 3.5.4. Details of buddy Trusts will only be kept for contingency purposes should a fault be experienced.

Note

This links with Section 12 Management of Major Influxes of emergency calls

5. Contingencies and Review of EACR Numbers

EAs should prepare contingency arrangements to cover the receipt of emergency calls during conditions of serious breakdown either in the ECN or the EA communications system. These arrangements must be planned to cover every control centre which normally receives emergency calls and should be tested periodically to ensure that both CHAs and EA staff are familiar with them. The EAs should make use of automatic call diversion facilities where possible.

CHAs and the EAs should regularly monitor the efficiency of the 999/112 emergency call arrangements and arrange regular liaison meetings at both national and local levels.

The EAs should inform the 999/112 liaison contacts points in all CHA’s of all changes to any EACR connect-to numbers giving a minimum of two weeks' notice. In the case of EA geographical boundary changes, at least one months’ notice should be given along with full written details of the changes. The date and time of all such changes should also be given.

The CHAs 999/112 liaison contact points should also give the EAs a minimum of two weeks' notice if their own contact numbers change. This information should be notified direct to the EAs.

6. Call Queuing Systems at EACRs

6.1 Queue Set Up

If the CHA operator is connected to a call queuing system, by clearing the original connection they are only placing themselves further back in the queue of calls and worsening their chances of being answered. To avoid this eventuality, the EA must notify the 999/112 liaison contact point in all the CHAs when planning to introduce or change a call queuing system. This is so the CHAs can amend their information to advise operators a queue is in use.

If a number is known to enter a queue CHA operators will be instructed to try it for 2 minutes before setting-up to the secondary and/or alternative number. The CHAs need to know which of primary, secondary and alternative numbers are in a queue, whether the separate numbers enter the same queue, and whether the separate numbers all have the same priority within the queue.

If the primary and secondary numbers are in a common queue, the secondary should be given priority in that queue, thereby recognising that a call attempt has already been queued for 2 minutes and keeps its queue position while trying a different line.

6.2 Queue Announcement/Message

The provision of a short acknowledgement message, or distinctive answer signal/tone, must be used by the EAs to inform both operators and callers that their calls are being held in a queue.

The queuing announcement/acknowledgement message should normally only be heard after at least 10 seconds of ringing tone, so that most callers would not hear it.

The announcement needs to terminate as soon as an EACR call taker becomes available and is presented with the call.

The recommended text for an announcement is as follows: - “You are connected to the Police/Fire/Ambulance Coastguard Service for emergency calls. We are currently receiving a high volume of calls. Please hold the line to report your emergency and you will be connected as soon as possible.” Where this is not appropriate, e.g., when reporting a fire, any announcement will have to be thought out very carefully.

Following an announcement, ringing tone should again be provided and the announcement repeated at regular intervals until the call is answered.

If it is not possible to use an announcement, then use of a distinctive tone to indicate that the call has been placed in a queue at the EACR needs to be used. This could either be a variation to the normal public network ringing tone, or a periodic interruption of normal ringing tone (being provided by the EACR switch) by a different burst of a distinct tone.

Note

If possible, and the queuing system allows, extra information can be temporarily inserted into an announcement to try to manage demand and caller expectations.

For example: "……..We are currently receiving a high volume of calls due to (major incident details) ……."

7. Call Monitoring and Disconnection

Once the EA answers, the CHA EO will (unless interrupted by the caller) give the name of their Operator Centre (to facilitate recall), the caller's number and advise if it is a payphone or mobile caller. They will also advise the EA if:

- the caller's number is not known.
- the caller is from Limited Service State
- it is a Telematics call
- it is a call from an international EA Control Room
- Virtual Assistant
- VOIP from an unconfirmed location

CPs are required to provide location information to the emergency services for 999/112 calls and this is made available to EAs via the BT CHA's location "hub", known as the EISEC hub (Enhanced Information Service for Emergency Calls). To allow EAs to access locations, the caller's number and the Operator Centre identity are automatically transferred to the EACR's switch and do not need to be verbally passed, however, verbal handover will still be given in the circumstances listed in the bullets above.

EAs are then expected to automatically request location information from the EISEC hub – this is civic address (including postcode) for fixed lines, and a location circle for mobile locations, of varying radius dependent on technology used to determine location (can be a less than 10 metre radius if satellite positioning available).

Note

Guidance on use of the location provided by EISEC is contained in Appendix C.

If the caller remains anonymous the EA should request the customer identification details once the call has been terminated. Note this will not be possible in the case of Limited Service State calls.

Once connected to an EACR, the CHA EO will listen to establish that the caller is able to pass details to the EA call taker. Once the EO is satisfied that the caller is responding to the EA's questions, they will release the call from their position, to be held on the Operator Centre switch. However, if the EA's EISEC access is unavailable, and the caller is very young or very old, is panicking, is having difficulty speaking or where it is apparent that English is not the caller's first language, then the EO will normally listen throughout as it is very likely they will be asked for location information. If further 999/112 calls require answering, operators may make themselves available to answer the new calls, thus removing themselves from the listening process at an earlier stage.

7.1 EA Requires Further Information

If a caller clears before all relevant information is obtained, and the EA Controller does not end the call, the CHA will be alerted and an EO will return to the line for assistance. Calls are generally only released from the CHA when both parties using fixed lines clear.

If necessary, the EA can call the CHA Operator Centre national contact points using the agreed contact numbers. The EA will need to quote the time of the call, the caller's telephone number and whether the call was Limited Service State.

If the call has not been released, the operator centre may then be able to associate the 999/112 call with the operator concerned who can then be advised to go back into circuit to speak further with the EA. If the call has already been released, then the call records can be quickly consulted to confirm the basic call details.

The CHAs will hold call records for a period of 3 years. Call records include basic details such as callers telephone number, the time and date of the call, the EACR telephone number to which the call was connected and indications of any delay in connection or any unusual aspects of the call. For fixed calls the installation address will normally be present too. Note in the case of Limited Service State calls there will be no valid CLI for the caller.

Note

Call recording access information is in Section 13

8. Calls Without Service Request

There are many calls to 999/112 where a caller does not actually request an EA. In the overwhelming majority of cases these are children playing or customer misdials, but there is always a possibility of it being a genuine caller who cannot speak. In order to prevent the Police from being overwhelmed with these calls, the National Police Chiefs' Council (NPCC) have requested that the CHA operators filter such calls. To do so the CHAs use carefully designed procedures agreed with the NPCC.

8.1 Calls Without Service Request from Mobile Phones

Very large numbers of accidental 999/112 calls are received from mobile phones. CHA EOs try to obtain a response by asking the normal questions – for example "which service is required", and "if you cannot speak but need help please tap the handset screen, cough or make a noise." In cases where there is:

- no speech and nothing apart from general noise is heard;
- or where the voice link is terminated during CHA questioning, perhaps with background speech or noise;

It is recognised that there is a negligible chance that it is genuine and the CHA EO can end the call.

Where there is no response, the voice channel remains open and background voices are present, it is recognised that the CHA cannot decide whether an EA is needed. In this case the call is connected to a Police voice response system hosted by the Metropolitan Police Service which asks caller to press "5" twice if help is required. If 55 is pressed then immediate connection with the appropriate Police authority is made.

For any cases where suspicious noises are heard the CHA can override the above procedures and connect to a Police EACR.

8.2 112 Calls Without Service Request from Fixed Phones

Switches on fixed networks still have to recognise loop/disconnect dialing and 112 calls can therefore be readily generated by faults within networks or customer equipment. Despite the use of network filters to prevent such calls reaching the CHA EOs, many are still received and appear as silent, open lines, or noisy lines with crackling and interference sounds. The overwhelming majority of callers on fixed networks use 999.

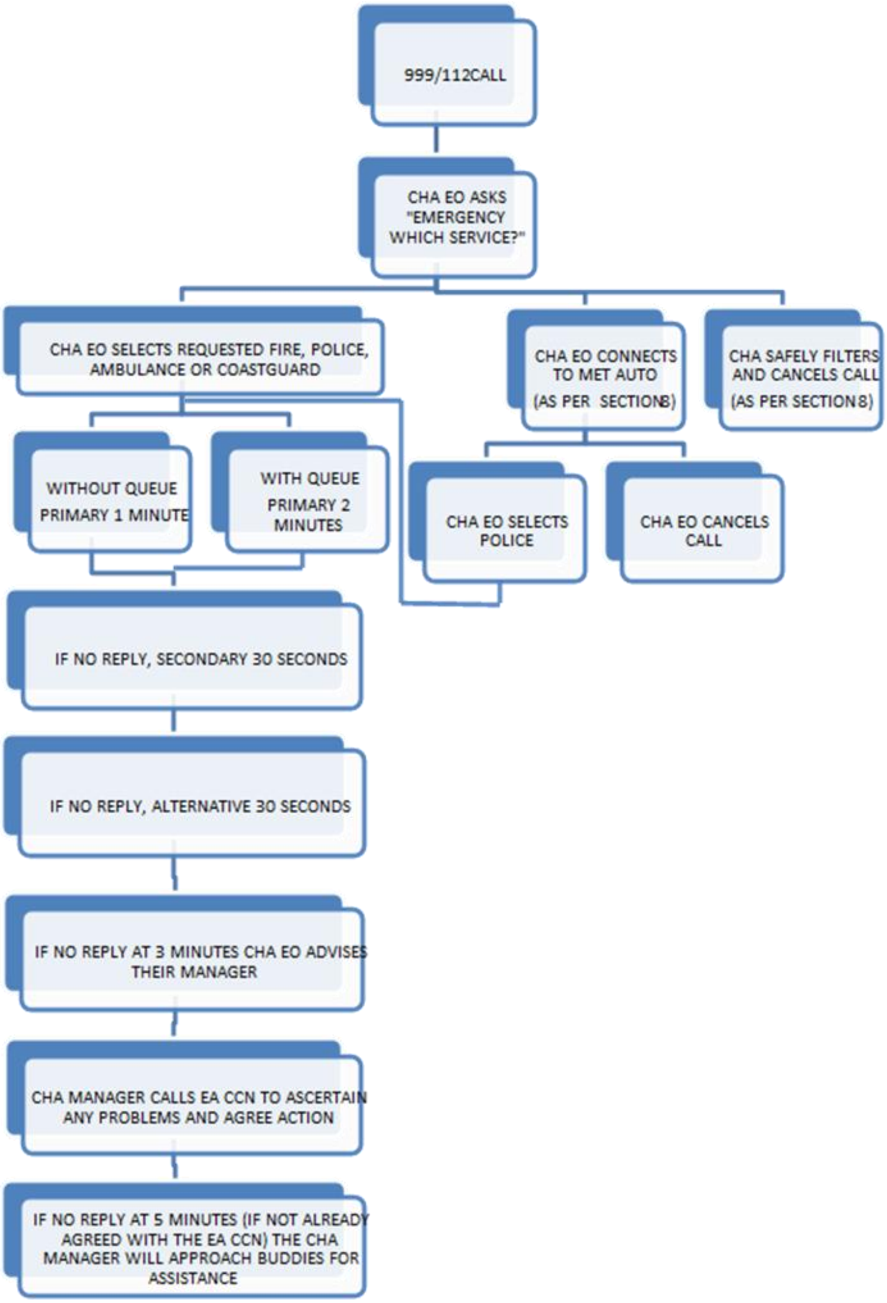
Once CHA EOs have asked their normal questions without response, it is agreed that there is an almost negligible risk of missing a genuine call. Therefore, for noisy 112 calls the CHA EO can end the call. Silent 112 calls will be connected to the Police voice response unit at New

Scotland Yard. This provides one final check as described above to cover the rare case of a genuine 112 caller unable to speak being on the line.

For rare noisy calls to 112 on which the level of noise makes it impossible to listen, the call's voice link is cleared before details are passed to the Police by CHA EOs.

This procedure will be kept under review in light of genuine use of the 112 code. This issue should decrease in light of the transition to all-IP.

8.3 999 Call Connection Flow Chart



Note

The flow chart above no longer reflects the timings of connections to the primary, secondary and alternative numbers for Ambulance calls due to their use of the Intelligent Routing Platform (“IRP”). Please see Chapter 3.5.4 for more information.

9. Post-call Provision of Call and Caller Information

9.1 Processes

The CHAs will pass information to the EA about the call and caller as the call is transferred to the EA.

If EISEC is used EAs can rapidly obtain fixed line installation addresses and approximate mobile locations for calling numbers by accessing the secure database that temporarily holds the details for 999/112 calls.

> **Note**
>
> This information is not available if the call was made using a mobile phone in Limited Service State (refer to Section 15.1).

CHAs do endeavour to maintain a high degree of accuracy but are not able to guarantee the accuracy of the (fixed line) name and address information supplied verbally or via EISEC. CHAs do not accept any responsibility for the accuracy of the information provided to them by the CPs and have no liability for any injury, damage or loss caused by the provision of the information. Appendix D contains practical background and guidelines for the use of (fixed line) name and address information.

EA requests for information as described in this Section must be kept to a minimum, consistent with the essential nature of emergency call needs.

9.2 The Emergency Period

The ‘emergency period’ is defined as within one hour of the termination of the emergency call. Within the emergency period the EA can re-contact the CHA and/or contact the relevant CP in order to:

- seek confirmation or reiteration of the call and location information
- obtain information about a caller (subscriber information)
- seek updated location information (if available).

A description of services and capabilities available at each CP can be found in the ‘Dropped 999/112 and Disconnection procedures Crib sheet’ which is circulated to all Force Control Rooms on behalf of the 999 Liaison Committee

Specific guidance concerning the circumstances where data may be acquired or disclosed in relation to 999/112 calls is laid out in Chapter 10.5 of the Investigatory Powers Act 2016 Communications Data Code of Practice. See Appendix F of this document.

As a rule, the acquisition and disclosure of communications data (subscriber information, call details and location information etc.) must only take place under an IPA authorisation or notice.

To enable the provision of emergency assistance in response to emergency calls which are ‘dropped’ or incomplete, Chapter 10.5 (Public Emergency Call Service 999/112 calls) allows an EA to call upon a CHA or relevant CP to disclose data about the maker of an emergency call within the emergency period outside the provisions of the Act (IPA). It goes on to say that in practice this means sufficient detail to identify the origin of the call and, if appropriate, to enable the deployment of an emergency service to the scene of an emergency.

After the emergency period an EA must contact an accredited IPA Single Point of Contact (“SPoC”) in order for the SPoC to follow their processes to submit an IPA authorisation or notice for the communications data required. In practice an EA could also be an IPA SPoC but the principal remains, outside of the emergency hour acquisition and disclosure of communications data must only take place under an IPA authorisation or notice.

Note

An extract from the IPA (Acquisition and Disclosure of Communications Data) Code of Practice can be found at Appendix F.

9.3 Confidential Customer Information

All caller and caller location information provided by the CHAs and CPs is strictly confidential and must be kept and used securely by the EA and only used for the purposes of dealing with emergency incidents. CHAs and CPs will authenticate EAs requesting information and will check the validity of requests.

9.4 Mobile Phone Subscriber Checks

Mobile phones making emergency calls could fall into one of a number of categories:

- Contract (or post-pay) account: Generally, these are accounts where customers pay for their mobile phone usage in arrears and are required to register (and have verified) their name, address and other contact details.
- Pre-pay account: These are accounts where usage is paid for in advance. Customers are not required to register their name, address and other contact details although some do. Registered contact details are generally not verified.

- Foreign roamer: These are (contract or pre-pay) customers from a mobile phone network operating in a foreign country which has a roaming agreement with one or more of the UK mobile phone networks. The UK network CPs do not have access to any subscriber information for these customers.
- Limited Service State (LSS) call: These are (contract or pre-pay) customers who have no coverage from their home (UK) mobile phone network but are able to make 999/112 calls in LSS using another UK mobile phone network. See Section 14 for further details.

9.5 Mobile Virtual Network Operator (MVNO)

These are sometimes called ‘resellers’ and operate ‘virtual’ mobile phone networks using some or all of the elements of the networks of the main four mobile phone CPs: EE, Three, Vodafone and VM-O2.

Arrangements for the provision of MVNO subscriber information varies from MVNO to MVNO. Refer to Dropped 999/112 and Disconnection processes’ for further information.

Where a mobile subscriber is owned by an MVNO rather than the CP, it may be the responsibility of that MVNO to supply subscriber information. Where required an MVNO should ensure that it has a process to pass relevant subscriber information to an EA as required. This process should be available on a 24 hour basis.

10. Misrouted Calls and Mobile Zone Problems

It is possible that the postcode, national calling number, zone code display or Cell ID could give an incorrect but apparently valid EA routing to the emergency operator. This could occur as a result of:

i. a faulty console;
ii. a fault in transmitting the information from the database;
iii. more commonly, where a mobile handset has accessed a base station located in a different zone or Cell to the one where that handset is actually located. Handsets automatically search for the strongest signal and connect to the base station that they find provides it – this does occasionally mean it is not necessarily physically nearest to the site of the incident being reported. (This most typically will occur across river estuaries.)
iv. where a call has originated from a private network that extends over several areas (DDI systems). In these cases, the number automatically presented to the CHA EO can be different to the number that the caller provides when asked by either the CHA or the EA. This is because any 999/112 call can be fed into the ECN in one of the areas. This can lead to problems as the CHA EO will be presented with a telephone number applicable to this area, and therefore routed accordingly. This will only become apparent at the EACR when the caller is questioned as to their location.
v. where a call is an Limited Service State roamer. In this case, a calling line number ("CLI") may or may not appear but in any case, it will not correspond to the caller's number.

The CHA EO connects using the automatically presented information, as indicating the most appropriate EACR and to minimise any delay in connection. It is the responsibility of the EA controller to establish that the call is proper to the EA area and to instigate means of transfer if it is not. This can be achieved in a number of different ways in decreasing order of preference.

i. Taking details of the call and passing the information to the correct EACR;

If option (i) is not possible, then

ii. Recalling the CHA operator and requesting that the call is passed to another EA or County. This may not be possible if the CHA EO has already relinquished the call into the system.

If it is not possible for the EA to advise on the correct alternative EA or county, then the CHA EO will instigate a mobile call trace procedure as described in Section 10.2 to identify where the mobile handset accessed the mobile network. The CHA EO will then re-route the mobile customer to the applicable EA control.

Note

Call traces can only be carried out providing the mobile customer continues to hold the connection.

The EA should not advise the caller to redial 999 and request a specific area or EA. The CHA EO will automatically connect the call to the relevant EA as per the presented routing information and override the request from the caller.

10.1 Boundary Overlap (Mobile ECNs)

Where cell or exchange coverage areas straddle two or more EA coverage areas, the CPs (or their call handling agents) will nominate an Emergency Authority Control Room (in consultation with the EAs as appropriate) to which 999/112 calls from that cell/exchange are to be directed. If this choice later turns out to be incorrect, the arrangements can be changed by mutual agreement. Comparison of exchange area/cell site boundaries and EA boundaries shows that exchange/cell boundaries are unlikely to overlap more than two adjacent EA coverage areas.

10.2 Failure to Display Zone Or Cell Id Information (Mobile ECNs)

Failure to display zone or Cell ID information is extremely unlikely. However, should this occur, the CHA EO will ask the caller for location information (e.g. county) and route the call accordingly.

If the location is not known, the CHA EO will tell the caller that there is a network fault and that some information checking will be necessary. The ECN will use inherent network facilities to locate the cell of origin. If the ECN is unable immediately to determine the zone code or Cell ID it will persist in determining the caller's location.

Note

It is recognised that extra time will be needed to go through this procedure.

It is essential that the ECN gives the CHA Operator a zone identity or Cell ID on all occasions and with the minimum of delay.

The cell and zone location facilities described above may be possible providing the calling mobile holds the connection. If the caller clears, traces are not generally possible, but records are kept by the CPs which include time of call, duration, originating number, the cell which received the call and the resulting zone code (where applicable). These records are kept and will be readily available for cross-checking for approximately three months. Currently only EE and Vodafone keep these call records. Both VMO2 and Three no longer do so and will not be able to cross check.

Note

In the case of Limited Service State roamers, the data will be restricted to time, duration, the Cell Id and the IMSI number which identifies the SIM used and records may not be kept in these cases.

11. Routing Methods and Fault Management

11.1 Routing Methods and Network Security

In the Telecommunications network, the standard method of routing 999/112 calls from CHA EOs to EACRs will be via a CP provided communications path. Calls will normally be connected over the primary route to the nominated EACR. The Telecommunications network provides good transmission and fast call set-up. It also provides forwarded 999/112 calls with automatic alternative routing through the network to the nominated EACR, along with priority treatment over normal calls, which particularly helps on occasions when the network is busy. This method of routing provides sufficient resilience to meet the obligations under the General Conditions of Entitlement.

All mobile CPs will provide capacity from their 2G/3G Mobile Switching Centres (“MSC”), or 4G/5G Emergency-Call Session Control Function (“E-CSCF”) to enable calls to be routed to the appropriate first choice operator centre. The capacity provided will be secured by supplying diverse transmission routes, where possible, to minimise the effect of equipment or line plant failures.

Alternative routings from the MSCs / E-CSCFs will be automatically invoked when route congestion to, or failure at, a switch is detected.

Note

Any circuit suspected of being faulty will be removed from service until such time as engineering tests have proved the fault rectified and operational tests have been performed.

All UK operators have plans to or in the process of planning for sunset their 2G/3G networks and in doing so have to make sure that Emergency calls still work during the migration and post migration to 4G/5G This includes operator coverage plans and device support

Further safeguards exist in the form of a back-up service provided between the Operator Centre switches which will be effective against temporary closure of the Operator Centre site due to emergencies such as fire alarms or bomb threats. BT’s CHA currently has 6 OC sites and 3 primary network sites.

11.2 Priority Fault Repair Service

EACRs need to make clear to the CPs that provide their 999/112 service that priority should be given to maintaining service to all 999/112 circuits in accordance with the relevant regulations.

Condition A3 of Ofcom’s General Conditions of Entitlement states that all CPs must take all necessary measures to ensure uninterrupted access to Emergency Organisations as part of any Publicly Available Telephone Services offered. This will involve the protection of the physical and functional operation of such systems and services against malfunctions or failure caused by electrical conditions, signaling protocols or call volumes.

11.3 Isolation of Customers Due to Network Failure

In the event of a major failure to a part of an ECN, the CP will notify the affected EAs as soon as possible after the failure is identified or anticipated. CHAs and CPs will assist the EAs in order to overcome the situation. A copy of the BT CP process has been circulated to all EA contacts. If the 999 call handling system is impacted, the CHA will advise all Emergency Services via email to the agreed 24/7 contacts.

EAs and relevant CHAs should adopt local contingency arrangements for 999/112 call handling in the event of a network failure. EAs should prepare contingency plans to cover lack of network access to 999 through their Local Resilience Forums.

11.4 Draft Media Statement for CHAs/Cps and EAs

Please see Chapter 14 for all advice on communication to the public. Advice Comms lines.

12. Management Of Major Influxes of Emergency Calls

12.1 Advance Warning of Major Incidents

In some cases, prior warning of major incidents is possible. Extreme weather, special events such as the World Cup, and Christmas / New Year social activity are all well-known sources of additional 999 calls and are known about in advance.

All EAs and CHAs need to assess the Impact of Weather Forecasts available from the Met Office and the Environment Agency/ Scottish Environment Protection Agency.

Staffing arrangements for both EA and CHA control rooms should be adjusted accordingly in the light of these forecasts and known impacts from previous similar events. CHAs and EAs should exchange information on predicted impact on the 999 service of predicted events, for example the volumes/times of increased 999 demand. In particular an EA should notify the CHAs as soon as possible if a likely staffing shortage is identified that could affect handling of 999 calls.

The use of media announcements should be considered by EAs in order to minimise public reliance on the 999 system. Advice could be given about preparations already in hand for their area, what will be considered an emergency, where to seek further guidance or assistance etc.

12.2 Contacts

EAs should check with CHAs to ensure they hold the correct “buddy” arrangements, offering additional buddies if necessary, e.g. more distant ones likely to be unaffected by the event. Agreement should be confirmed between buddy EAs on how information will be passed e.g. email, electronic interfaces, CAD/Command and Control logs, radio, DEIT, MAIT, or dedicated voice channels etc. Consideration should also be given to the use of non-emergency numbers (such as 111 or 101) where appropriate.

Conference Calls should be considered where ongoing contacts are going to be needed during a major incident and if many parties are going to be involved at the same time. These can be established between EAs and CHAs or simply among EAs with feedback/instruction for the CHAs. Such Conference Calls should follow an established process and can be part of any Silver Command structure established to manage an event or established specifically to manage 999 call handling issues.

12.3 Modifications to Queueing Systems

See Section 6 for general use of queues. If possible, and the queuing system allows, extra information can be temporarily inserted into the announcement to try to manage demand and caller expectations. For example: "……..We are currently receiving a high volume of calls due to (major incident details). ……..”

12.4 Changes to Call Handling Procedures

It is important that EAs and CHAs establish contact with each other as soon as possible after a major incident/event has been identified that is affecting 999. This may be before the impact is fully assessed.

The EA will contact the CHA to advise of the impact they are expecting/experiencing and give revised call connection instructions if a change is required.

The CHA will contact the EA if the EA’s time to answer is exceeding 3 minutes (i.e. when neighbouring services would already be assisting on multiple calls).

In such circumstances the phrases below would normally be agreed between the EA and CHA for the duration of the incident/event in question.

If contact is not quickly possible, so as to protect calls to other services, the CHAs would move automatically to these expressions as soon as (a) delays of greater than 3 minutes in connection to one or more EAs are sustained for more than 15 minutes, or (b) the delays are such that they severely affect the CHA's own ability to answer 999 calls quickly for more than 15 minutes.

Expression: “(EA name) is very busy with calls relating to (major incident details) at (location). I will try to connect you.”

12.5 Temporary Procedure Contingency

Temporary procedure contingency is designed to maintain the 999 service standards during any critical period of high 999/112 call volumes or call answering difficulties where the CHA experiences a delay in answer by EA’s. This can be agreed with individual EA’s, used for a type of EA or for all EA’s, where there is a queue announcement in place.

When temporary procedure is used, if a verbal handover is not required and callers are not distressed, then once a call is set up and ringing out to the EA concerned, and providing that the call is in a queue, the caller may be told; “You are in the queue for (EA Name), please wait for the (EA Name) to answer. Can I leave you in the queue?” As long as the caller does not disagree, the operator will release the call into the system and go on to answer the next emergency call.

12.6 OPERATION WILLOW BECK

Operation Willow Beck is a procedure developed in partnership with BT, to manage the distribution of exceptionally high emergency call volumes to the fire and rescue service. The following information is extracted from "NFCC Operation Willow Beck Procedure" (Version 1.2 01.12.2023, Section 2.2, 2.3 and 2.4)) from the National Fire Chiefs Council.

12.6.1 OPERATION WILLOW BECK Procedure

Fire and rescue services who participate in Operation Willow Beck, will enter into an agreement for the proportion of overflow calls they are willing to accept on behalf of other fire controls. This will enable the redistribution of 999 calls, relating to the spate or spike condition, to assisting fire controls who have entered the agreement and are able to participate.

Operation Willow Beck consists of two options:

- Operation Willow Beck - Redirection
- Operation Willow Beck - Filtering

Both options will result in calls being connected to assisting fire controls. However, when filtering is requested, some calls will be filtered out meaning the caller will be asked to ring back if their situation worsens. These calls will not be connected to a fire control.

12.6.2 OPERATION WILLOW BECK - Redirection:

- Business as usual calls will be connected to the affected fire control
- Calls relating to the spate or spike condition will be connected to one of the assisting fire controls

12.6.3 OPERATION WILLOW BECK – Filtering

- Business as usual calls will be connected to the affected fire control
- Calls relating to the spate or spike condition, and where life is in danger, will be connected to one of the assisting fire controls
- Calls relating to the spate or spike conditions, where no life is in danger, will be asked to ring back if their situation worsens. These calls are filtered out and will not be connected to a fire control.

For both Operation Willow Beck Redirection and Operation Willow Beck Filtering the fire control must inform BT of the exact location and nature of the incident.

Once initiated, the call handling agent should connect calls as follows:

- Route 1 – calls not relating to the current spate condition will be connected to the affected fire control, for example calls for fires, road traffic collisions and other special services

- Route 2 – calls relating to the current spate condition will be connected to one of the assisting fire controls in Operation Willow Beck

It should be noted that there may be occasions where the call handling agent connects Route 2 calls via Route 1.

If Operation Willow Beck with filtering has been requested, calls relating to the current spate condition and where life is not in danger, will be advised to ring back if their situation worsens

If the call handling agent is in any doubt as to the safety of the caller, they will connect the call to an assisting fire control and the caller will not be asked to ring back. This may lead to many calls being connected to assisting controls.

The number of calls an assisting fire control may receive will be based on the proportion initially agreed. This proportion may be temporarily increased if one or more controls temporarily opt out of Operation Willow Beck. Should an FRS decide to permanently opt out, their proportion of calls will be distributed, by agreement, to one or more of the remaining fire controls.

12.6.4 OPERATION WILLOW BECK- Higher Priority Incidents

Higher priority incidents are those which require an operational response, (including life risk incidents). Life risk incidents are those where a person is subject to immediate danger of serious injury or death due to the situation or environment they are in.

On receipt of a higher priority or life risk incident the assisting fire control should return the details to the affected fire control by the nominated telephone number or the Airwave fire hailing group. MAIT may be used where available and where the affected fire control indicates they can receive incident information this way.

12.6.5 OPERATION WILLOW BECK - Lower Priority Incidents

Lower priority incidents are those which may not require an operational response (including non-life risk incidents).Non-life risk incidents are those where it is determined that a person is not in immediate danger, property is not at risk and the call does not require an attendance.

On receipt of a lower priority or non-life risk incident, the assisting control should return the details to the affected fire control by email. MAIT may be used where available and where the affected fire control indicates they can receive incident information this way.

12.7 Flooding and Major Incidents

If specifically requested by a Fire Service trying to prioritise many flood or Major Incident related calls and allow normal fire service calls to be answered, the CHA may agree to ask further simple questions before connection, in an effort to filter-out non-emergency cases on behalf of the EA.

First question would be: “Are you reporting a flood/Fire/Wildfire in [location – down to street/area]

If the incident is not related, the CHA would connect straightaway as normal. If a flood or major incident, a further question would be asked: “Is anyone in danger?” If the answer was “yes”, or doubt existed, or the caller insists on connection, then the call would be connected to the Fire Service as normal. If not, the caller would be told “XXXX Fire Service is very busy with flood/major incident related incidents. If the situation becomes worse, please call back.”

This instruction from the Fire Service would need written confirmation (email) from the Fire Service to the CHA to back-up a verbal instruction.

If contact from the Fire Service is not quickly possible, so as to protect calls to other services, the CHA’s would move automatically to these expressions as soon as

Delays of greater than 3 minutes in connection to one or more EA’s are sustained for more than 15 minutes, or

The delays are such that they severely affect the CHA’s own ability to answer 999 calls quickly for more than 15 minutes

Note

A ‘Major Incident’ in this case is from the definition from the JESIP Joint Doctrine, is; “An event or situation with a range of serious consequences which requires special arrangements to be implemented by one or more emergency responder agency”

12.8 Media Announcements During a Major Incident

Please see Chapter 14 for all Public Advice Comms lines.

13. Recording of Calls

CHAs will record all calls terminating on 999/112 circuits. Calls are recorded from the time the call is answered by BT's stage 1 PSAP Operator until the caller clears and the circuit is released, regardless of whether the CHA has monitored throughout. The original 999/112 recording tapes are kept by CHA's for a period of 3 years.

To access a recording from BT, a Court Order or Customer and EA consent is required. Police should sent via the Single Point of Contact (SPoC) via control room. SPoCs should be contacted first for advice and guidance. As Fire, Ambulance and Coastguard often do not have an agreed SPoC, they should contact the BT Disclosure team.

The following levels of authority are required for access to be given:

- Police – Assistant Chief Constable
- Fire – Assistant Chief Officer
- Ambulance – Director of Operations
- Coastguard – Chief Coastguard

CHAs should apply to the Chief Officer of the relevant EA for similar recordings of calls made by the EAs.

CHAs inform their customers that they should use the 999/112 number when making emergency calls. CHAs do not record emergency calls made to the 100 number (general operator services access code). However, such calls are handled despite the use of the incorrect number.

14. Guidance on Public Advice

This guidance was developed by the Crisis Communications team within the Government Communication Service, with input from behavioural scientists and in coordination with lead government departments, emergency services, and the Liaison Committee. These guidelines are intended to provide direction for emergency services and their lead government departments during outages or periods of high demand.

14.1 Broad Principles of Messaging

Lead Government Departments and emergency authorities tasked with communicating to the public should adopt the Krebs approach in their delivery of key information. The important principles are to communicate consistently and frequently; and to set expectations that the information may change as more information becomes available. Public advice should look to outline the following:

What to communicate

- Tell the public what is known.
- Tell the public what is not known.
- Tell the public what actions the government is taking, and why (this may include actions to mitigate the crisis, and actions to reduce uncertainty).
- Tell the public what they should do, and why.
- Tell the public when to expect more information.

14.2 Channels of Communications

For significant outages lasting an appreciable amount of time it may be appropriate to post an update via social media. Where possible, this should be agreed or communicated with the relevant emergency authorities, BT and other communication providers, to ensure consistency in messaging. A follow up status message should be provided as soon as there is an update or when an issue is closed.

14.3 Suggested Lines to the Public During Outage Scenarios

The below lines have been signed off following consultation with HMG, EAs and the 999 Liaison Committee.

14.3.1 National 999 System Failure Scenario

e.g. June 2023 whole system outage

England, Scotland, Wales and NI	
Ambulance, Police, Fire and Rescue Service and Coastguard	We are aware of an issue with the 999 call system currently. A full investigation is underway to resolve this as quickly as possible. Members of the public with genuine emergencies (for instance if there is an immediate threat to life or property) should take the following actions: • try to call 999 or 112 in the normal way as you may still be able to get through. • If however you are having difficulties connecting, you may contact 101 for police or 111 for health (please note the 111 service is not in operation in Northern Ireland). • You can also report non-emergency police matters online in England, Wales and Northern Ireland on www.police.uk. In Scotland, you can use https://www.scotland.police.uk/. • For non-urgent health queries, you can use the 111 service online: https://111.nhs.uk/ We will give you an update on the situation as soon as we have more information. ****These lines are for an outage, were it to happen tomorrow. Further work is being done by ops colleagues to address the lack of an alternative number for Fire and Rescue/Coastguard services.***
Suggested social media posts	We are aware of difficulties in using the 999 service and are working quickly to resolve this. If you cannot connect to 999 or 112, you may contact 101 for police or 111 for health. Non-emergencies can be reported online. Please look out for further updates. ****Northern Ireland to omit 111 line in the tweet.***

14.3.2 High Demand Scenario

Eg. Heatwave 2022, whole system increase influx of calls

England, Scotland, Wales and NI	
Ambulance, Police, Fire and Rescue Service and Coastguard	The emergency services are currently experiencing a high demand of 999 calls due to .... [insert lines on emergency causing this level of high demand] The emergency services (police, fire, ambulance and coastguard services) and the 999 call handling agents are doing all they can to respond to this unusually high level of demand. We would appeal to the public to consider whether or not it is necessary to dial 999. If your call is an emergency, for example, any life-threatening situation, a fire, or a crime in progress, you should continue to call 999 or 112. But we would like to remind you that if your call is for a non-emergency situation there are alternatives to the 999/112 system. Police: 101 Ambulance service: 111 You can also report an issue which is not an emergency to the police in England, Wales and Northern Ireland by going online here: www.police.uk. In Scotland, you can use: https://www.scotland.police.uk/. For non-urgent health queries, you can use the 111 service online: https://111.nhs.uk/ We will provide an update on the situation as soon as we have more information.
Suggested social media posts	The emergency services are currently experiencing a high demand of 999 calls due to (insert issue). If you cannot get through to 999/112, please call 101 for police or 111 for health. Non-emergencies can be reported online. Please look out for further updates.

	Northern Ireland to omit 111 line in the tweet.

14.3.3 LOCAL OUTAGE SCENARIO

England, Scotland, Wales and NI	
Ambulance, Police, Fire and Rescue Service and Coastguard	When a network failure requires a statement to be issued to the media, the following lines should be considered. The **below is not a prescribed set of lines**, as there is likely to be variability in the scale of the emergency and differing levels of capacity across emergency authorities. The below lines can be modified where appropriate, however we advise you to follow the Krebs method, as explained above. **Please select the below bullets as applicable to your circumstances.** • (Communication Provider / Call Handling Agent name) have notified us of problems affecting telephone service in the xxx area. They are currently working to restore *full* service to everyone as soon as possible. • If you have an emergency, you can try calling 999/112 from a mobile phone as the mobile network may still be working. If it isn't working you may still be able to make an emergency call via another mobile network. • We are asking people to avoid making non-urgent calls to 999 for the time being so that all the available phone lines can be used for real emergencies (for instance If there is an immediate threat to life or property). • If you are reporting an issue to the police, which is not an emergency, in England, Wales and Northern Ireland you can go online here: www.police.uk. For Scotland, you can access https://www.scotland.police.uk/. • For non-urgent health queries, you can use the 111 service online: https://111.nhs.uk/ • Extra police patrols are out on the streets, so if you need help and are able to get out, try to get the attention of a police officer or a passing police car.

	• You could also seek help by going to your nearest police station, hospital, fire or ambulance station in person. • We will give you an update on the situation as soon as we have more information.
Suggested social media post	You may face problems calling 999 from a landline in the xxx area, try calling from a mobile phone as mobile networks may be/are working normally. Normal service will be restored as soon as possible. Non-emergencies can be reported online. Please look out for further updates.

14.3.4 PARTIAL OR FULL MOBILE NETWORK OPERATOR (MNO) *OUTAGE SCENARIO*

England, Scotland, Wales and NI	
Ambulance, Police, Fire and Rescue Service and Coastguard	To avoid public confusion, Mobile Network Operators (MNOs) should initially communicate outage details and issues connecting to 999 directly via their own media teams. When a network failure requires a statement to be issued to the media, the following lines should be considered. The **below is not a prescribed set of lines**, as there is likely to be variability in the scale of the emergency and differing levels of capacity across emergency authorities. The below lines can be modified where appropriate, however we advise you to follow the Krebs method, as explained above. **Please select the below bullets as applicable to your circumstances.** • (Communication Provider) has notified us of problems affecting their network. They are currently working to restore *full* service to everyone as soon as possible. • In an emergency, continue to call 999/112 as you should be able to

	make an emergency call via another mobile network, or, use a landline if available. • If you are reporting an issue to the police, which is not an emergency, in England, Wales and Northern Ireland you can go online here: www.police.uk. For Scotland, you can access https://www.scotland.police.uk/. • For non-urgent health queries, you can use the 111 service online: https://111.nhs.uk/ • We will give you an update on the situation as soon as we have more information.
Suggested social media post	You may face problems calling 999 from (XX) network, please continue to call 999/112 from a mobile phone as you should be able to make an emergency call via another mobile network or use a landline. Normal service will be restored as soon as possible. Non-emergencies can be reported online. Please look out for further updates.

14.3.5 COMPLETE OUTAGE

- In the case of a national power outage

England, Scotland, Wales and NI	
Ambulance, Police, Fire and Rescue Service and Coastguard	The scale of a national power outage and its impacts across the United Kingdom are largely unknown. Similar to the suggested lines in the local outage scenario, the below advice provides a baseline for communicating with the public, and can be **tailored by departments/ emergency authorities - with the Krebs method in mind.**

	• In an emergency try to call 999 or 112. If you cannot call 999 or 112, try to flag down any emergency service vehicle. Even if it is not the emergency service you require, the emergency vehicle will be able to provide you with assistance and can contact the service you need. • We are asking people to avoid making non-urgent calls to 999 for the time being so that all the available phone lines can be used for real emergencies (for instance If there is an immediate threat to life or property). • Extra police patrols are out on the streets, so if you need help and are able to get out, try to get the attention of a police officer or a passing police car. • You could also seek help by going to your nearest police station, hospital, fire or ambulance station in person. • You should try to get to a hospital with an emergency department (as this is more likely to have power). • As power is restored, 999 may become busy or still be experiencing disruption - if you need help, keep trying until you get through. You may also contact 101 for police or 111 for health (please note the 111 service is not in operation in Northern Ireland). • We will give you an update on the situation as soon as we have more information. The public should also listen to the radio for regular broadcasts (you may wish to add that BBC radio 2 and 4 are still functioning once you receive confirmation that these stations are operating).
Suggested Social Media Post (if social media is running)	You may be facing problems calling 999 due to a power outage. We're working to restore the service. If you have an emergency, please flag emergency service vehicles on the streets or go to your nearest police, fire station or hospital. More updates to follow.

15. Other Types Of Calls

Other types of call may include:

- Limited Service State
- Relay UK
- 999 BSL (Emergency Video Relay Service)
- eSMS
- Telematics
- eCall
- Other Vehicle Activations
- Third Party Call Centre Emergency Calls
- Google 'Loss of Pulse'

15.1 Limited Service State Emergency Calls

This section contains details of 999 or 112 emergency calls made from mobile telephones which do not have network coverage from their own provider. These calls are referred to within the Mobile CPs as "Limited Service State".

From 2009 Ofcom expected all UK mobile network operators to introduce the ability for users to make calls to 999 or 112 on another mobile network – thereby allowing users to make such calls when out of the coverage of their home network and thereby maximise the ability for users to obtain help quickly. This functionality should not be confused with foreign mobiles visiting the UK as roaming agreements would be in force that would allow the user to make and receive calls as normal.

15.1.1 Considerations For The Emergency Services

Whilst Limited Service State has the benefit of allowing users to dial 999 when outside of their home network coverage, it does present challenges for the emergency services. It is important to note that these types of calls will not contain all of the data normally associated with a conventional emergency call and there will be limitations with regard to locating where the call was made from.

- The call will present to the CHA operator with either no telephone number (CLI) or a default number. Either way the caller's own mobile number will not be presented.
- Location for these calls is confined to either the Zone code or in some cases the Cell ID which doesn't identify the location precise enough to confidently dispatch assistance. These codes are used to route calls to BT and the emergency services only and therefore do not provide any assistance to identify the caller's location.
- The CHA may be able to pass the caller's telephone number verbally if the caller relays it while waiting for the EA to answer.
- The CHA will advise the EA that the call is a "Limited Service State" call.
- If contact with the caller is lost, it will not be possible for the CHA or EA to contact the caller even if the mobile number has been obtained UNLESS the caller has moved and is now within coverage of their home network.

- If the call drops, the EA would normally contact the CHA to gather additional information. In such circumstances, the EO should advise the CHA that the call is a Limited Service state call. This will alert the CHA to the processes they have in place to assist with trying to trace such calls. (The EO should quote Limited Service State caller together with the date, time, number which was displayed on EISEC and the customer telephone number (if it was obtained.)
- It may be possible for MNOs to provide additional information. However, this could, in some circumstances take up to two days due to the technical challenges locating such a call although this will vary between networks. It is generally not possible for any of the MNOs to provide "real time" information within the golden hour.
- A mobile telephone number will not be available to the MNO who carried the call. The only information that will be presented to the MNO is likely to be the IMSI.
- Further work would therefore be required by the EA to identify the "home" network in order to acquire subscriber details.

A disadvantage of Limited Service State is that users can make hoax calls to the Emergency Services on another network to their own even if the user has been disconnected from their home network. EAs should therefore be aware that requesting users to be disconnected for making hoax calls does NOT prevent them from making further calls as calls can continue as a Limited State Service call. This therefore makes the situation worse as the mobile number will no longer be presented at the time of the call.

As mentioned above, it may be possible for MNOs to provide additional information which would assist in identifying the caller but such an activity cannot be performed in real time and would require an IPA request to be issued to the MNO.

15.2 RELAY UK (Previously Next Generation Text (NGT)/Text Relay/Typetalk)

The Relay UK service is provided by BT on behalf of the communications sector for customers and end-users who are deaf, hard of hearing, deaf-blind, hearing or speech-impaired and use a dedicated app or a textphone to make and receive telephone calls.

15.2.1 The Relay Uk App And Textphone

Customers who cannot hear or speak clearly - or at all, can download the Relay UK app to their mobile, tablet or computer, link this to their phone number, and then make and receive telephone calls. Users can type what they want to say and read the responses within the app, with the call being facilitated by a Relay Assistant, or they can communicate directly with anyone else who's using the same app or a textphone. A traditional textphone can still be used on the service and is in essence a standard fixed line telephone but with the handset replaced by a keyboard and display.

15.2.2 Emergency Relay Uk Calls

As well as supporting standard phone calls, Relay UK also enables emergency calls to be made via the unique access code of 18000. There is no need to register or add ‘999’ or ‘112’ after this access code; all the caller has to do is dial 18000 and the call will be answered by a CHA Emergency Operator (EO), and a Relay Assistant will join the call at the same time. The Relay Assistant will relay the conversation by reading and speaking the typed text from the app or textphone user to the CHA EO, who will then connect the call to the correct EA. The CHA EO will then release the call so that the conversation continues between the caller and the EA, with the Relay Assistant typing what the EA says back to the caller. Some callers may speak instead of typing what they want to say, then the Relay Assistant will type back what the EA Controller says in reply, so the deaf or hard of hearing person can read what is being said. If the caller is speech impaired, they will usually type what they want to say for the Relay Assistant to speak, then hear what is being said in reply. Customers using the service will be connected in text to 18000 via the Relay UK system.

- The CHA EO will announce to the emergency services that the call is from a Relay UK user.
- The name and address details will be passed to the EA in the same way as for a voice 999 call.

If for any reason the Relay service is extremely busy a call may be answered by the CHA EO without the Relay Assistant being in conference. These calls will be handled in the same way as a silent call.

Visit the Relay UK website for more information - Contact 999 using Relay UK - How to use Relay UK | Relay UK (bt.com)

15.3 999 British Sign Language Emergency Video Relay Service

The 999 British Sign Language (999 BSL) Emergency Video Relay service can be used by any customer who’s first or preferred language is British Sign Language, and as an alternative to using the voice 999/112 or 18000 Emergency Relay UK services. The overall objective is to provide a customer experience as similar to that of the voice 999/112 access as possible. All UK Providers of fixed and mobile telephony services and providers of internet access services are required by regulation to provide, or contract to provide, a free 24/7 video relay service for British Sign Language (BSL) communicators, to contact the emergency services.

15.3.1 999 BSL App and Web Access

The 999 BSL app can be downloaded from the respective app stores for both iOS and Android mobile devices. The service can also be accessed via the web at www.999bsl.co.uk. The 999 BSL app and website allow BSL users to click on a single button to make an emergency call and be connected to a British Sign Language interpreter working remotely. The interpreter then calls the CHA EO (via a dedicated number), who handles the call in a similar way to a voice 999/112 call, connecting through to the requested EA, with the Interpreter relaying the conversation between the parties in real time. The app also enables a call-back option; this means that the emergency authorities can call a BSL user. The call will connect directly to the

Ofcom Approved Provider (currently Sign Language Interactions) call centre, where a BSL interpreter will answer and connect to the BSL user in a matter of seconds.

15.4 EmergencySMS (eSMS)

Guidelines for the eSMS service were established by the UK Government's 999 Liaison Committee and an MoU was drawn up to set out the service parameters, a copy of which can be found as Appendix G.

The SMS Emergency (eSMS) service is provided for people who are deaf, hard of hearing or speech impaired as an alternative option for contacting the EAs. The overall objective is to provide a customer experience as similar to that of the voice 999/112 service as possible.

Note

Due to the nature of how SMS works, calls will take longer than an 18000 or 999/112 call – therefore eSMS should only be used where there's no other option.

An eSMS can be sent from a standard mobile handset, but end-users need to register to use the service by texting the word 'register' to 999. When an eSMS is received, the originating phone number is automatically checked to ensure that it is registered, before it is translated and passed on to Relay UK. An eSMS from an unregistered phone number will be rejected and sent an automated message advising that the user is not registered and needs to contact the emergency services using other methods.
Caller Location information is made available from the originating network and provided to the EA in the same way as for voice emergency calls. The Relay Assistant will connect to the CHA EO, who then connects to the correct EA as soon as possible once the emergency service is known - and to the Police if no EA is requested.

An outgoing emergency voice call will be generated using the eSMS originating phone number and will get directed to the CHA EO. Responses from the EO or EA Controller to the Relay Assistant will be converted into an SMS by the eSMS service and sent back to the originating mobile phone number.

Follow up arrangements for EAs on calls are through the same contact points as for emergency voice calls.

More details on eSMS can be found on the website – www.relayuk.bt.com/emergencysms

15.5 Third Party Call Centre Emergency Calls

A process can be agreed with the CHA to ensure calls are routed to the correct area Emergency Service and any available location data is made available through EISEC. Please see section 15.7.1 for vehicle activations via third party call centres.

15.6 eCall And Telematics Emergency Calls

eCall based on 112 is a mandated system (with Telecommunications Regulation and Vehicle Type Approval rules) that provides an automated message to the emergency services following a road crash which includes the precise crash location. The in-vehicle eCall is an emergency call (an E112 wireless call) generated either manually by the vehicle occupants by pushing a button or automatically via activation of in-vehicle sensors after a crash. When activated, the in-vehicle eCall device will establish an emergency call carrying both voice and data (using a modem) directly to the designated 112/999 Public Safety Answering Point, PSAP. The voice call enables vehicle occupants to communicate with the trained eCall operator. At the same time, a minimum set of data will be sent to the eCall operator receiving the voice call. The minimum set of data contains information about the incident including time, precise location, vehicle identification, eCall status (as a minimum, indication if eCall has been manually or automatically triggered) and information about a possible service provider.

Telematics is a service where mobile phone technology in a vehicle is linked with position information derived from satellites. The service pre-dates eCall and uses an SMS or mobile data message to transmit vehicle and location data prior to setting up a voice call to a specified number (not an emergency).

The CHA EO (designated PSAP) will connect the call to the appropriate EA in the normal manner but highlight that the call is a “Telematics mobile” or “eCall mobile” and add GPS location information or, if there is a delay in receipt of location, will interrupt with location details when available. The map reference is a fully numeric 14 figure Ordnance Survey grid reference. Satellite positioning of the eCall system now provides an accuracy of +/- 15 metre from the final location provided. The CHA EO is able to do this as in addition to receiving the voice call they will also receive data containing the vehicle and location details from a modem transmission at start of voice call (for eCall) or via a separate data message (for Telematics calls). The location data and vehicle data is provided to EAs over EISEC. This type of call normally includes voice and data, however, on occasion the CHA EO may receive a voice only call or a data only call.

The EA operator is then able to directly question the caller as normal.

If the caller is unable to supply the needed information, and it is not present on EISEC, the EA operator can ask the EO for further details and usually obtain direction of travel (N, NW, W, etc or degrees), make, model, colour and registration of vehicle or VIN number, and sometimes other personal information supplied by the customer.

Limited Service State calls cannot be made over the Telematics Service. The service requires connection to the home network to function.

The full Telematics Protocol for handling eCalls and Telematics emergency calls can be found at Appendix G.

15.7 Other Vehicle Activations

Ford has an in-car system which allows the customer's mobile phone to be connected to an in-car monitoring system. In the event that the car is involved in a crash where the air-bag is deployed or the fuel pump has been cut off, the car will make an emergency call from the customers mobile. This system has now been superseded by the mandatory introduction of eCall based on 112. When the CHA answers the call the in-car system will play a message to the EO giving the co-ordinates of where the caller is. If the call is silent, or there are problems obtaining location from the caller, the call playback facility can be used to allow the co-ordinates to be played directly to the EA operator.

15.7.1 Vehicle Emergency Activations Using 3rd Party Call Centre

Some car manufacturers have chosen to have their in-car emergency call activations (eCall and Telematics) answered by a call centre, which can be located in the UK or elsewhere. The call centre will establish if the customer requires emergency assistance in the UK and if so, they will connect the customer to the CHA in the UK. The third-party call centre will ask for the emergency service type required and pass all relevant information including the location and vehicle details prior to opening a conference so that the EO can speak directly to the customer. The data generated by the vehicle can be passed verbally or by electronic transfer.

15.8 Google 'Loss of Pulse'

Loss of Pulse Detection is a feature on the Google Pixel Watch which can place a call to 999 if a loss of pulse is detected. Further information about the feature can be found at

What is Loss of Pulse Detection on my Google Pixel Watch? - Google Pixel Watch Help

These calls will be identified by the EO by the automated message played on the emergency call and will be connected to Ambulance. The calls will be dealt with as a critical call by the CHA if there is any delay in answer by the Ambulance.

APPENDIX

A) GLOSSARY OF TERMS
B) VERSION CONTROL
C) GUIDANCE ON USE OF LOCATION INFORMATION
D) BEST PRACTICE GUIDE FOR CPs ON NAME & ADDRESS INFORMATION (LANDLINES)
E) MOBILE DISCONNECTION
F) COMMUNICATIONS DATA CODE OF PRACTICE (PURSUANT TO IPA)
G) eSMS MEMORANDUM OF UNDERSTANDING FOR PEOPLE WHO ARE DEAF, HARD OF HEARING or WHO HAVE IMPAIRED SPEECH
H) TELEMATICS PROTOCOL

APPENDIX A: GLOSSARY OF TERMS

Term	Description
999/112 Liaison Committee	Committee facilitated by the Department for Science Innovation and Technology which brings together Emergency Authorities, Government Departments, Call Handling Agents and Communications Providers to consider matters of common interest.
AML	Advanced Mobile Location
BAPCO	British APCO
BT	British Telecommunications plc
Caller Location	Any data or information processed in an Electronic Communications Network (**ECN**) which gives the geographic position of the terminal equipment of the person making the emergency call.
CAD	Computer Aided Dispatch
CCN	Critical Contact Number
CHA	Call Handling Agent
CLI	Calling Line Identity
COCOM	Committee on Communications
CP	Communications Provider (or **TO**) is an organisation that provides an **ECN** or an **ECS**. This term is used within the Communications Act 2003.
DDI	Direct Dial In
DEIT	Direct Electronic Information Transfer
EA	Emergency Authority
EACR	Emergency Authority Control Room
ECN	Electronic Communications Network
ECO	European Communications Office
ECS	Electronic Communications Service
EISEC	Enhanced Information Service for Emergency Calls
Emergency Caller	A 999/112 caller
Emergency Roamer	A 999/112 caller using a network that is not their home network – also called Limited Service State (**LSS**)

EO	Emergency Operator
eSMS	Emergency Short Message Service
IMSI IPA	International Mobile Subscriber Identity Investigatory Powers Act 2016
JESIP	Joint Emergency Service Interoperability Programme
LSS	Limited Service State
MAIT	Multi Agency Information Transfer
MET AUTO	Voice response system hosted by the Metropolitan Police Service
MNO	Mobile network operator
MSC	Mobile Switching Centre
MVNO	Mobile Virtual Network Operator
NPCC	National Police Chiefs' Council (formerly **ACPO)**
OC	Operator Centre [Fixed Network]
PCA5	Percentage of Calls Answered (by CHA) within 5 seconds
PECS	Public Emergency Call Service
PSTN	Public Switched Telephone Network
SIM TO	Subscriber Identity Module Telecommunications Operator – synonymous with **CP.** This term is used within the Investigatory Powers Act 2016**.**
URN	Unique Reference Number
VoIP	Voice over Internet Protocol
XD	Ex-directory

APPENDIX B: VERSION CONTROL

Control Information	
Author/Editor:	DSIT Telecoms Resilience

Change History

Version	Date	Changes	Prepared by
1.0	**Feb 2025**	**Version 1**	**Telecoms Resilience and BT 999**

Distribution
Communication Providers, BT the Call Handling Agent, the Emergency Authorities and their
Copy for information
999 Liaison Committee, Home Office, Department for Health and Social Care, Department for Transport, Department for Science Innovation and Technology.

APPENDIX C: GUIDANCE ON USE OF LOCATION INFORMATION

Although CHAs endeavour to ensure data accuracy they cannot claim 100% accuracy for the address information which is supplied to them by the CPs serving the end-users. Experience over the years with addresses supplied verbally by operators, and now automatically by EISEC or similar systems, shows a high degree of accuracy is maintained on the CHA databases.

The following guidelines will help call takers/despatchers when working with caller name and address information.

EA Call Taker/Dispatcher Guidelines

1. Fixed Lines

In the first instance the EACR call-taker can have a high degree of confidence in the accuracy of the CHA-provided address.
There are known instances where the address provided by the customer may well differ from that provided by the CHA. In these known cases, the CHA will provide indicators with the data to show that the address provided may not be accurate. These situations arise because of the communications technology the caller is using:

1.1. Private Networks

Many businesses have private networks known as Private Branch Exchanges (PBXs).
For example, a company could have a published main number of 0113 237 6400 with 30 lines that can be used for incoming and outgoing calls for the 200 employees, each with their own extension. External customers dial the main number and are connected to the required extension by a switchboard operator. Outgoing calls go automatically through the switchboard, usually by dialling a network access code such as “9”.

Other PBXs provide Direct Dial In (DDI) facilities, which for this example may lead to the company having 200 incoming numbers (e.g. 0113 237 6400 to 0113 237 6599), all accessible directly to external callers. However outgoing calls still use the main number 0113 237 6400. This is the number displayed to the CHA’s agents and passed to EACRs. The address automatically presented to an EACR is the one held by the CHA that is associated with this main switchboard number. Since some companies spread their DDI extensions over a number of sites, or across a large site, the caller may give significantly different address details to those supplied by the CHA as well as passing their own DDI number which will be different from the presented, main number. Such PBX/DDI addresses are marked with **EXT** at the front of the name to show that the caller may well not be at the main address but that of an extension on a different site.

Even if the EACR calls the CHA to check address details for the customer-passed DDI number, the CHA’s records will only show the address of the main switchboard, as customers do not in the main provide their CPs (and hence the CHAs) with addresses for individual extension numbers. The only way to check a caller-passed location on such a call is to recall the DDI extension or main number and enquire of the customer or their switchboard staff respectively.

1.2. VoIP – Voice over Internet Protocol - Calls

There are different types of calls made over VoIP networks, some may be from products where the call can only be made from a fixed location (such as the BT 'Digital Voice' product) whilst others may allow their customers to be able to make calls from different locations. This is referred to as a 'nomadic' service.

There may be different levels of 'nomadicity' for numbers as well depending on the network product in service.

Where there is any element of 'nomadicity' for a number, if the CP providing service has advised the CHA of this, the address will be marked with ***EXT*** to warn that the caller may not be at the location given.

There are also VoIP Service Providers that do not allocate individual CLIs to every registered user; this may be because the Service Provider provides some of their customers with an outgoing calls only service where it is not possible to call the user back, for example some of SKYPE's customers. In these cases, a default, non-geographic CLI is presented on all emergency calls.

Currently it is not technically feasible to provide accurate location details for Nomadic VoIP users other than the default address; however it may be possible for the Service Providers to trace the originating IP address and block hoax callers, or provide the registered user's contact details.

1.3. Other reasons when the address provided by the caller may not match that provided by the CHA:

(i) The caller has recently moved address: Name and address details are updated as necessary between once and up to four times a day depending on the CP. There are thousands of updates made each day involving new customers taking out service or existing customers moving address and taking their telephone number with them. The names and addresses provide by the CHA for these numbers will be incorrect for about half a day and is a known source of discrepancies between reality and data the CHA holds.

(ii) The caller is reporting someone else's emergency: There are instances where an emergency incident is reported by a third party and where the address they report may not match that held by the CHA.
Alarm monitoring companies provide a commercial service to care homes, other businesses and individuals. They will receive a call from a client and then contact the EACR on their behalf. The telephone number they quote to the CHA as being that of the original caller and the address they provide, possibly from a confused caller or from their own database, are subject to increased errors.
Neighbours and bystanders can provide address information for affected premises which they think is correct but may well not be and which will not relate to the telephone number of the person making the call
Non-emergency services such as 101 or 111 may take calls which then need escalating to a 999/112 call. The call will usually present to the EACR in the same way as that from an alarm monitoring centre

2 Mobile Calls

EACR need to ensure the full EISEC specification** is supported by suppliers of their control room systems – this allows both an approximate network location and a more precise handset location where available (known as Advanced Mobile Location, AML) to be retrieved through the EISEC interface (separated by a short time delay to allow handset to fix its location). It also allows retrieval of location and vehicle details for eCalls and Telematics calls.

EACRs need to display full location information on a call taker's mapping screen – this will be a location circle of varying radius for mobile callers (showing both a network location and a handset location where possible), and not simply a point as seen for fixed line addresses. The diagram below illustrates this, showing the differing precision of network and handset locations. There will also be a Level of Confidence value that caller is within the circle that should be displayed.

In addition, command and control software needs to be able to use the mobile location which is provided using a map reference, not a civic address with a postcode.
EACRs should ensure that call handling systems allow call takers to match the coordinates provided as part of the handset locations to (a) nearby locations used within local databases that can be relayed verbally if needed, or (b) matched directly to responding resources with a free text description added if more appropriate, e.g. '100 metres west of {Nearest Property, or Road Junction, as matched by local address database}'.

There are some cases where no network location can be provided – this is the case with Limited Service State calls (also known as emergency roamers) where a handset has no coverage from its home network but can use another network, or from overseas users in the UK roaming onto UK networks. AML is also not available from Limited Service State calls.

Handset Location (AML) is available from most smartphones, but is not always technically feasible (e.g. if no access to satellite positioning), or possible from other handsets without a smartphone capability. At the time of this update (October 2018) it is present for almost 70% of mobile emergency calls.

3 When things go wrong

3.1 For Fixed Lines

Despite the data integrity checks carried out and the annual audit of the data held by the CHAs, things can go wrong and the address provided by the CHA for a 999/112 call can be incorrect.

If the EACR call-taker has reason to doubt the address provided by the CHA (from what the caller has said) then they can recall the Operator Centre in the normal manner and ask for the address to be checked. The CHA will then check their own database and if necessary, will either contact the serving CP to confirm the caller's address or will provide the EACR with a contact point in the CP that the EACR can deal with directly.
Any discrepancies found when using EISEC or similar systems should be e-mailed to the CHA's data discrepancy team in accordance with the detailed procedures provided by the CHA..

3.2 For Mobile Calls

A network or handset location is sometimes not possible due to technical limitations or systems issues, and there can be occasional errors in network or handset information used to derive locations Verbal confirmation with callers wherever possible, as well as comparing network and handset locations, helps to quickly identify any such issues.

APPENDIX D: BEST PRACTICE GUIDE FOR CPs ON NAME & ADDRESS INFORMATION (LANDLINES)

This set of guidelines has been developed through discussions with Ofcom to help ensure Communications Providers (CPs) achieve compliance with the General Conditions of Entitlement (GC A3.5 and GC A3.6).

They provide reassurance for the Emergency Authorities on the efforts made to ensure that address information supplied to them by the CHA on each 999/112 call is as correct as practically possible.

1. The Communications Providers will be expected to ensure that the location information provided to the Call Handling Agent (CHA) is correct and up to date. The CP shall take steps to ensure that installation addresses are not confused with billing (or other) addresses related to the end user.
2. New location information and routine updates to location information should be notified to the CHA by the CP within one working day of the new, or changed, line termination arrangements being completed. With regard to ported lines, both the gaining and losing CPs need to comply with this recommendation to ensure the integrity of the data held by the CHA
3. The CP should ensure that addresses supplied are validated by reference to a recent (not older than 6 months) version of the Postcode Address File (PAF) and that it also conforms to the data structure required by the CHA.
4. The CP should ensure that the CHA is notified of newly activated, or re-assigned, number ranges before records of CLIs and addresses for those CLIs are sent to the CHA. Failure to do so will delay the loading of the data onto the CHA's database as the CHA carried out further data integrity checks
5. The CP should, at a minimum, conduct an annual audit to compare names and installation addresses held on the CP's own systems with the location information held on the CHA database. Ad hoc audits may be required in cases where the level of discrepancies for a CP begins to give cause for concern or where significant system changes have taken place.
6. The CHA will normally inform the CP of any discrepancies or missing data within one working day of these being identified by the CHA operator or within one working day of these being communicated to the CHA by an Emergency Authority (EA). Such discrepancies will be notified to a designated point of contact within the CP.
7. Once a CP has been notified of a discrepancy, the corrected information should be supplied to the CHA within two working days. If there has been no correction from the CP after five working days, the CHA will escalate the issue to its senior contact at the CP.
8. The CHA should keep records of discrepancies in location information supplied by CPs for at least 12 months to assist Ofcom to monitor compliance with GC 4.2.
9. The CP should provide a contact point for the CHA to be able to urgently verify names and addresses against the CP's own records for calls where an EA needs assistance. This contact point should be available 24 hours a day, 365 days a year.
10. If several CPs are involved in calls reaching the CHA, for example where a CP re-sells another CP's services, then all CPs concerned need to co-operate so that appropriate arrangements for the CLI and associated name and address records to be sent to the CHA are in place and that the above guidelines can be observed.
11. A CP who has exported a number should also transfer the ownership of the associated Emergency Services address record to the importing CP's CUPID. The CP who is now

providing service to the end-user should also request to take ownership of the associated Emergency Services address record from the exporting CP's CUPID. These transactions should be placed at the end of the next working day after the number has been successfully exported.

Private Networks of Large Enterprises

- As a minimum, the address provided by CPs to CHAs should be the address of the point of access from private network to the PSTN. (Often the headquarters or largest site for the Enterprise.) In such cases, the CP should apply an "uncertainty flag" to the address records for the relevant CLIs to alert emergency call handlers to confirm address information verbally with the end user (caller)
- Ensure the Enterprise Customer (Large Corporate customer) understands the limitations for emergency location information from all customer sites.
- CP to advise* the Customer of options** that may allow site-specific location information to be delivered for each access point within its Wide Area Network (WAN) to the 112/999 CHA operators.
- Where a CP has the facility to provide such an option, the CP shall recommend that its customer register and update the location information with it (with accompanying contractual terms). If CPs cannot rely on Enterprise Customer updating the CP, then address records should again be marked with uncertainty flag.

* Where known CP has a reasonable expectation, or has been informed by the Enterprise, that the service is to be accessed from several locations.
** For example, any options recommended by the NICC **http://www.niccstandards.org.uk/**

APPENDIX E: MOBILE DISCONNECTION

MOBILE TELEPHONE DISCONNECTION REQUEST FORM	
TO	(network)
LOG NUMBER	
Mobile Number ___________________________ has made ______ hoax/malicious emergency calls to ______________ (authority name) at the following dates and times ………….……………………	
We have so far been unable to find the owner of the handset. If we wish to take further action against the owner, we will request any further information through the formal process under IPA.	
DESCRIPTION OF CALL(S) Please describe in detail what the calls were and, if appropriate, what services were despatched.	
Please consider voluntary disconnection of this number for breach of the Communications Act 2003 and your airtime terms and conditions.	
This Authority recognises that by disconnecting the mobile number the caller can continue to make calls to the emergency services on a different network (Limited State Service) which means that the phone number will no longer be presented. The impact of this has been taken into account and this Authority accepts the risk on behalf of all emergency services.	
REQUESTERS DETAILS	
NAME	
RANK	
SIGNATURE	
DATE	
APPROVERS DETAILS	
NAME	
RANK	
SIGNATURE	
DATE	

APPENDIX F: COMMUNICATIONS DATA CODE OF PRACTICE (PURSUANT TO IPA)

Extract from: Investigatory Powers Act 2016 Communications Data Code of Practice

Chapter 10 SPECIAL RULES ON THE GRANTING OF AUTHORISATIONS AND GIVING OF NOTICES IN SPECIFIC MATTERS OF PUBLIC INTEREST

Public Emergency Call Service (999/112 Calls)

10.8 It is usual for telecommunications operators to disclose, at the time of such a call, some identity (caller line identity) and caller location information data (fixed or mobile, if available) to the emergency services in order to facilitate a rapid response to the emergency call.

10.9 Telecommunications operators should take steps to assure themselves of the accuracy of the information they may be called upon to disclose. Any known limitations in this accuracy, particularly for location, should be proactively disclosed to the emergency services. Emergency services should be aware that communications data may not always be available for disclosure by the telecommunications operator depending on the particulars of the communications service used to make the call.

10.10 If the emergency service control room has reason to doubt the address provided for a fixed-line number by the emergency operator (from what the caller has said) then they can contact the Operator Centre in the normal manner and ask for the address to be checked.

10.11 The emergency service can call upon an emergency operator or relevant service provider to disclose data about the maker of an emergency call within the emergency period one hour from the termination of the 999/112 call.

10.12 It is appropriate for the emergency service or emergency operator to require the telecommunications operator to disclose any further caller location information that might indicate the location of the caller at the time of the emergency call. Within one hour of the 999/112 call, it is also appropriate for the telecommunications operator, acting in the belief that information might assist the emergency service to respond effectively or efficiently to the emergency, to proactively disclose to the emergency service or emergency operator any further information about the location of the caller at the time of the emergency call or a new location the caller has moved to, if it is within the one hour period.

10.13 If an emergency call is disconnected prematurely for any reason, technical or otherwise, and the emergency operator is aware or is made aware of this, then the emergency operator can elect to represent the data disclosed when the call was to the emergency service initially. This voluntary disclosure would fall outside the scope of the Act.

10.14 Some telecommunications operators have provided secure auditable communications data acquisition systems for the disclosure of communication data under the Act. Where these exist, it is appropriate for emergency services to be provided with accreditation details to use them for acquiring data about the maker of an emergency call or caller location information, as appropriate, only during the emergency period.

10.15 When a secure auditable system is not available, a manual application for data can be made. The Public Emergency Communications Service Code of Practice contains the process to be followed.

10.16 If the emergency call is clearly a hoax, there is no emergency. Where an emergency service concludes that an emergency call is a hoax and the reason for acquiring data in relation to that call is to detect the crime of making a hoax call – and not to provide an emergency service – then the application process under the Act must be undertaken.

10.17 Should an emergency service require communications data relating to the making of any emergency call after the expiry of the emergency period of one hour from the termination of the call, that data must be acquired or obtained under the provisions of the Act.

10.18 Where communications data about a third party (other than the maker of an emergency call) is required to deal effectively with an emergency call, the emergency service may make an urgent oral application for the data. Disclosure of that data would also fall within under the provisions of the Act.

10.19 Increasingly, members of the public are using non-emergency numbers to request assistance. For instance, a caller might dial either 101 or 111 or other relevant services to seek non-emergency assistance). In some circumstances the call handler may consider it more appropriate that an emergency response is made for instance when the health of the enquirer suddenly deteriorates or a suspect returns unexpectedly to the scene of a crime. In such circumstances the one-hour emergency period and related provisions detailed above apply, even though the number dialled was not an emergency number.

10.20 The Act does not seek to regulate either the actions of the call handler or the provision of data by the telecommunications operator.

APPENDIX G: eSMS MEMORANDUM OF UNDERSTANDING FOR PEOPLE WHO ARE DEAF, HARD OF HEARING or WHO HAVE IMPAIRED SPEECH

1 Introduction and Scope

This Memorandum of Understanding (MoU) is formed between the organisations, set out in section 2 below, necessary to deliver emergency SMS messages in the UK, made from any mobile communication device over Mobile Networks to the Emergency Authorities (EAs).

This SMS Emergency (eSMS) service is provided to improve emergency access for disabled people as an alternative option for contacting the EAs for those unable to use the voice 999/112 service and, in particular, for those with hearing or speaking difficulties who routinely use SMS and either do not use Relay UK or may be in circumstances where it is not available.

This MoU covers the arrangements for the communications processes and interfaces involved in supporting this service when used in England, Wales, Scotland and Northern Ireland. It follows the guidelines established by the UK Government's 999 Liaison Committee, which held discussions between Emergency Authorities (EAs), Call Handling Agents (CHAs), the Mobile Networks, Ofcom, relevant Government Departments and representatives of groups representing deaf, hard of hearing and speech impaired people. The service set-out in the MoU was trialled from September 2009 through to April 2011 and results were regularly reviewed by a Steering Group chaired by the Department for Communities and Local Government (DCLG). The 999 Liaison Committee recognised in January 2012 that the Steering Group had ensured that the service was now operating in accord with this MoU and could be considered as fully launched and business as usual alongside the voice 999 service.

Provision of the service is mandatory for the mobile networks – General Condition C5.10 reads: 'Regulated Providers who are Mobile Service Providers must provide any End-User of their Mobile Communications Services who has hearing or speech impairments with Mobile SMS Access to Emergency Organisations by using the emergency call numbers "112" and "999" at no charge.'

2 Organisations Involved

- Mobile Network Operators (MNOs): The Party responsible for enabling the SMS messages to and from the caller to the eSMS service.
- SMS Aggregator: the Party responsible for receiving the data messages from the different MNOs, and passing them on to the modified Relay UK SMS interface for handling by the 999 Call Handling Agency and Relay UK.
- Relay UK: BT service that facilitates the use of voice connected services by deaf, hard of hearing, speech-impaired and deaf-blind, app and textphone users (textphones using ITU-T V.21 real-time text over voice path)

- Call Handling Agency (CHA): The Party responsible for the link to the Emergency Authorities. The UK CHA is BT.
- Ofcom: the independent UK communications regulator

- DCLG: UK Government Department providing Chair and secretariat for 999/112 Liaison Committee at time of the service's development and trial
- RNID: Royal National Institute for Deaf People (previously Action on Hearing Loss) which is registered charity 207720, the largest membership organisation representing deaf and hard of hearing people in the UK (and communicating with other groups representing deaf or hard of hearing people).
- Emergency Authorities (EA): These are the Police, Fire and Rescue, Ambulance and Coastguard services.

3 SMS Emergency Service - Overview

The overall objective is to provide:

- a single national access mechanism for SMS messages to the 999/112 service to assist those with hearing and speaking difficulties that make the use of the voice 999 service difficult
- direct access to all the emergency services who should not need to make substantial changes to their call handling equipment or operator training
- a customer experience as similar to that of the voice 999/112 access as possible
- access to the automated location systems that are available to voice, app and textphone calls

In addition:

- End-users will be required to register
- End-users will not be charged when using the service
- The service will be limited to end-users in the UK originating an SMS on UK networks and cannot cover foreign roamers on UK networks (due to SMS handling, and as they would not be registered)

3.1 Network Operators

The eSMS can be sent from a standard GSM or later generation mobile handset.

For some MNOs the eSMS can be sent only if the mobile phone is active and in credit (even though eSMS is not chargeable).

End-users will not be charged for sending the eSMS, or receiving messages associated with the original request

An eSMS originating from a handset registered for the service but roaming outside the UK mainland may be rejected

The SMS Centre (SMSC) will recognise the 999 and 112 emergency SMS code

An eSMS will be passed through a MNO's network to an SMS Aggregation Service in the normal way for each operator's network (e.g. SMPP interface)

The mobile network will allow their normal "message sent" responses but will not confirm successful delivery on networks where it is technically feasible to suppress such messages.

Location information will be provided within the limits of the existing interface for emergency voice calls when requested by the CHA, using the CLI of the handset originating an eSMS as well as sending Advanced Mobile Location (AML) data where available

The MNO will deliver the eSMS messages according to its business-as-usual SMS criteria.

The MNO will provide a 24 hour contact number for use by Relay UK, or the SMS Aggregator in the event of technical problems with SMSC failing to receive data messages. This route will be used to liaise over all operational problems.

3.2 SMS Aggregator

- The SMS Aggregator (SMSA) will process varying message formats forwarded from SMSCs and send messages in a standard format to the eSMS service
- The message sent to the eSMS service will include the originating handset's telephone number (Calling Line Identification) and an identifier for the originating NO's SMSC.
- The SMSA will be forwarded to the eSMS service as soon as technically feasible from receipt of the SMS from the MNO.
- The SMSA will provide a 24-hour telephone contact for use by the eSMS service to liaise over all operational problems.
- The SMSA will use reasonable endeavours to monitor all interface channels and processes to ensure that the SMS messages are being delivered to the eSMS service.
- The SMSA will process an agreed message format from the eSMS service and send messages in required formats to the appropriate SMSC for onward transmission as an SMS to end-user's handset
- SMSA will allow delivery reports to be requested, received and relayed

3.3 eSMS

- Upon receipt of an Emergency SMS, the eSMS system will verify that the originating phone number is registered with the service
- An SMS from an unregistered phone number will be automatically rejected, however, an automated message will be sent back advising that the user is not registered and needs to contact the emergency services using other methods
- The system will check message content from registered users and filter out as spam messages those not containing words from an extensive list of emergency keywords. An automated message will then be returned back to the sender, inviting them to specify the service required if they have a genuine emergency
- The system will contain a list of problem telephone numbers, that are barred from using the service and from which all eSMSs will automatically be rejected. The system will

allow the barred number and registered lists to be updated and edited by the CHA through a simple interface
- The system will provide the option to reject messages where the delay in delivery is over a preset level
- Rejection of an SMS due to barring may result in an SMS being sent to the originating phone number. This will be configurable and dependent on the rejection reason
- The eSMS system will translate the SMS text in whatever format is being used into V.21 textphone text and pass it on to Relay UK over an 18000 voice call (18000 being the standard code for V.21 textphone emergency calls in the UK)
- The originating network identity will be used to create standard Interconnect Identification (II) digits
- An outgoing emergency voice call will also be generated with the II digits appended to the emergency digits (999 or 112) as the called-party number
- The Calling Line Identification (CLI) for the outgoing emergency voice call will be that of the originating phone number used to generate the SMS.
- The outgoing emergency voice call will be directed to the designated 999 Call Handling Service.
- Responses from the 999 Service operator or EA operator conveyed by the Relay Assistant will be converted into an SMS by the eSMS service and sent back to the originating mobile phone number. SMSs sent to a 'caller' will indicate that the message has originated from the eSMS service by marking the originator as being 999 or 112 depending on the destination of the incoming SMS.
- Subsequent SMSs from the originating phone number will be directed to the same outgoing 18000 call
- Subsequent SMSs from the originating phone number presented after a contact has been released by the 999/EA operator will be marked to indicate previous contact within an adjustable time limit.
- All SMSs will be logged at the eSMS service and stored for 6 months in order to provide an audit trail
- The system will produce an auditable record of all messages received by the system and any resulting outgoing SMSs
- Users will be registered as part of the eSMS service and will be made aware of limitations at registration, prompted to make sure they have a response from an EA, and if not responded to by SMS quickly, then to look for alternative means of help.
- The eSMS system may queue received SMSs so that a predefined number of outgoing 18000 calls is not exceeded.

3.4 Relay UK 18000 Service

Relay UK will handle the 18000 calls generated by eSMS as standard 18000 calls.

3.5 CHA responsibilities

- The 999 Service will be able to obtain phone location information from the originating network using AML data and/or existing location interfaces queried by CLI

- The 999 Call Handling Service will use the OS map reference of the centre of the area provided by MNO (currently Base Station coverage area) in order to route the call to the appropriate Emergency Authority (EA).
- The location information will be provided to the EA in the same way used for voice emergency calls
- The CHA will set-up the voice call as soon as technically feasible after the requested EA is known, and to the Police if no EA is requested.
- Follow up arrangements for EAs on calls will be through the same contact points as for emergency voice calls

3.5.1 Barring

- BT will bar a CLI on request from an EA and where there have been 3 or more nuisance texts received within a 24 hour period and which don't require an emergency service.
- If the customer makes multiple requests for an emergency service or has been previously connected on several occasions for a non-emergency situation, then BT may in consultation with the emergency services agree to bar a problem repeat caller.
- Before asking BT to bar a CLI , the EA concerned will warn the caller that their number will be barred if they continue to use the service inappropriately
- Emergency Services can request barring either during a call or by contacting BT.
- Where a request is received to unbar a CLI, BT will refer the affected party to contact the EA that originally asked for the barring to be applied. The bar will then be removed where the request has been approved correctly and requested by the EA concerned.

Note

Barring will prevent the customer from being able to send SMS text messages to 999/112. All other functions on the phone will be unaffected.

When a caller is barred, we will send a text message to their phone telling the caller that they cannot use eSMS. This message reads: -

"Due to inappropriate use *insert EA eg. Nottingham Police* have stopped your mobile using emergencySMS. If you think this is an error, please contact the Relay UK Helpdesk on 0800 500888 (text line) or email relay.uk.helpdesk@bt.com"

When the caller sends a further message to 999 after they have been barred, they'll receive the following message: -

"**999 Automated Reply** Sorry your phone is no longer able to use the emergencySMS service. No emergency service has been alerted. See www.relayuk.bt.com/emergencysms"

3.5.2 15 Minute Call end process

Where a call has been continuing for 15 minutes or longer the Duty Manager/chargeship will ask the emergency services if they will take over the call by either texting the customer back from their own community texting service or by the use of a mobile phone in the control room.

3.6 Royal National Institute for Deaf People (previously Action on Hearing Loss)

Royal National Institute for Deaf People (RNID) will inform its members and the wider community of deaf and hard of hearing people in the UK about the existence and proper usage of the service: RNID will also publicise the service appropriately through other organisations that serve those unable to hear or speak or that act as umbrella groups for deaf and hard of hearing organisations in the UK. Feedback from deaf and hard of hearing people received by RNID will be conveyed to the other partners

4 Liability

Each of the parties involved in connecting the eSMS messages to EAs will make reasonable endeavours to convey eSMS messages in a timely and accurate manner but cannot guarantee delivery. It will be made clear to the potential user at the time of registration that delivery cannot be guaranteed.

5 General

Any changes to this MoU or related documents, including software upgrades, need 3 weeks' notice of intention to change followed by agreement between the affected parties on the correct implementation plan. All changes are to be backward compatible.

For further information regarding this document please send to Secretary 999/112 Liaison Committee.

6 Schedule of automated messages

- After reading ALL this message, SEND THE WORD 'YES' TO 999 TO COMPLETE YOUR REGISTRATION - otherwise your phone isn't registered. In an emergency, you will know your message has been received ONLY when you get a reply from an emergency service; until then try other methods. Further details are available at www.relayuk.bt.com/emergencysms.
- Your telephone number is registered with the emergencySMS Service. Please don't reply to this message. For more information go to www.relayuk.bt.com/emergencysms.
- You have not finished registering. After reading ALL this message, SEND THE WORD 'YES' TO 999 TO COMPLETE YOUR REGISTRATION - otherwise your phone isn't registered. In an emergency, you will know your message has been received ONLY when you get a reply from an emergency service; until then try other methods. Further details are available at www.relayuk.bt.com/emergencysms

- **999 Automated Reply** Your telephone number is already registered with the emergencySMS Service.
- **999 Automated Reply** Your registration with the emergencySMS service has been disabled.
- **999 Automated Reply** Sorry your phone is no longer able to use the emergencySMS service. No emergency service has been alerted. See www.relayuk.bt.com/emergencysms
- **999 Automated Reply** 999 received text but can't send help yet. ONLY REPLY IF YOU NEED HELP by texting the word Police, Fire, Ambulance or Coastguard to 999
- **999 Automated Reply** SMS to 999 is not available from your mobile provider. No emergency service has been alerted. Please contact an emergency service using alternative methods.
- **999 Automated Reply** You are not registered to use the emergencySMS Service. Please use other ways to contact the emergency services.
- **999 Automated Reply** Messages from outside the UK are not accepted by the emergencySMS service

7 Glossary of Terms

CHA Call Handling Agency

EA Emergency Authorities: the Ambulance, Fire and Rescue, Police and Coastguard services

GSM Global Standard for Mobile phones

MNO Mobile Network Operator

8 Basis on which eSMS is provided

Following a voluntary trial, emergency SMS was made mandatory in the UK in May 2011. It is required under Ofcom’s General Condition C5.10, as set out in Section 1 of this document.

This change was originally notified by way of a document entitled **Changes to the General Conditions and Universal Service Conditions (Implementing the revised EU Framework)**, Statement and Notification, 25 May 2011, inserting a new clause and consequential definitions of “Mobile SMS Access”, “Short Message” and “SMS”.

The General Conditions were further updated in May 2023 and may be read in full here: General Conditions of Entitlement - Ofcom

APPENDIX H: TELEMATICS PROTOCOL

Memorandum of Understanding: SOS-Alerts using UK GSM Networks

1 Introduction

This Memorandum of Understanding (MoU) is formed between the parties, set out in section 2 below, to cover what is necessary to deliver SOS-Alerts to the Emergency Authorities in the UK, made from In-Vehicle Systems (Telematics Units) typically comprising a GSM mobile SIM with a Global Positioning by Satellite (GPS) device and a signal/data processor. The In-Vehicle System (IVS) can be linked to airbag and other crash sensors to automatically trigger an SOS-Alert call as well as having a separate “SOS” button for manual activation of an SOS-Alert. This SOS-Alert service is supplementary to the existing 999/112 service and is not intended to replace it. It consists of a voice call with related data carrying the location information and other data from the vehicle – the data may be transferred through a separate data or SMS transaction, or associated with the voice call.

This MoU forms part of the Code of Practice for the Public Emergency Communication Service (PECS), and is approved by the Government’s 999 Liaison Committee to cover the communications processes and interfaces involved in supporting the above devices when used in England (inc. Isle of Wight), Wales, Scotland and Northern Ireland. It is a ‘living’ document that will be subject to amendment from time to time (Clause 5 below refers) currently managed by BT’s 999/112 Policy Group using version control methodology.

2 Voice Call and Data Message delivery

Organisations involved are: -

- Value Added Service Supplier (VASS): The Party responsible for the specification of the hardware, the marketing of the service and compliance with this document. For example, for Telematics units and IVS eCall units this may be a Motor Manufacturer.
- GSM Network Operator (NO): The Party responsible for enabling the GSM Voice call and the transport of the data message from the hardware to the Emergency Data Service Provider.
- Emergency Data Service Provider (EDSP): the Party responsible for receiving a separate data message from the NO, conducting any processing required before passing it on to the SOS Call Handling Agency. EDSPs include - Allianz, PSA/IMA (Peugeot Citroen), Teletrac Navman, WirelessCar.
- SOS Call Handling Agency (CHA): the Party responsible for matching the voice and the data message together and providing the link to the Emergency Authorities. BT is the only UK CHA for 999,112 and Telematics SOS calls, and the designated Public Safety Answering Point (PSAP) for the UK.
- Emergency Authorities (EA): these are the Police, Fire, Ambulance and Coastguard services.
- A Third Party Call Centre : for some vehicles, the voice call may be first routed to a specified, qualified organisation that extends genuine calls to the CHA.

3 Requirements for SOS-Alert Services.

The UK Government's 999 Liaison Committee chaired the discussions between the EAs, CHAs and the existing EDSPs in August 1998. These concluded that:-

- the UK EAs fully support a dedicated emergency call button for SOS-Alerts and do not support units which use a single button for breakdown assistance, information services and SOS-Alerts
- the voice call should be connected directly to the CHA and quickly reach the EA in line with standard, voice-only 999/112 calls

Note**:** from 2013 the 999 Liaison Committee is now allowing voice calls to use a Third Party Call Centre prior to CHA under conditions in section 4 of the MoU, the location data should be processed immediately and forwarded to the CHA the data available from SOS-Alert calls should always include the Ordnance Survey map reference. The data should also include the vehicle's make, model, colour, registration number, or VIN, direction of travel, trigger information giving an indication of whether the alert was generated manually or by a crash sensor (along with crash sensor details, e.g. airbag, rollover, front, back, etc). The registered subscriber's/consumer's name may also be made available.

Note: from 1 April 2018, the vehicle data can also be provided using the standardised Minimum Set of Data (MSD) for eCall.

These requirements lead to the following responsibilities for the organisations involved.

3.1 Value Added Service Supplier Responsibilities

- There should be a dedicated button, or a dedicated combination of buttons, for SOS-Alerts labelled 'SOS'. There should be suitable safeguards to prevent accidental calls.
- The voice call should reach the EA via the CHA in the same way as with standard, voice-only 999/112 calls,
- The hardware should send a separate data message as quickly as possible to the EDSP with
  I. the MSISDN* of the sending GSM device if using a data message, or rely on SIM details sent if using SMS (so that mobile network inserts MSISDN)
  II. position information in the agreed EDSP format
  III. information on whether SOS-Alert has been manually or automatically triggered.
  IV. information on the crash sensor details (airbag, roll over, front, back)

*With separate paths for vehicle information and voice it's the MSISDN that is used at the CHA to match the voice.

- Using reasonable endeavours and where appropriate, the following information should be collated, and supplied to the EDSP:
  - vehicle make, model, colour and registration number (or VIN)
  - registered consumer's/subscriber's name
- The hardware should minimise the delay between initiation of the data call and the initiation of the voice call (performance criteria for this to be agreed with hardware manufacturers but less than 2 seconds is desirable).

- If the IVS unit is not able to establish a voice connection to the CHA for any reason, as a fallback it will dial the default emergency call number 112 automatically.
- In the event of a critical system failure which would result in an inability to execute a 112-based eCall, manufacturers shall provide a warning to the occupants of the vehicle. It would be up to the vehicle owner to get the system fixed.
- For eCalls, the IVS provider is often the vehicle manufacturer (VM), but could be an after-market provider (AMP). The VM or AMP is responsible for ensuring the IVS complies with the relevant eCall standards – see Appendix 1. In particular the VM/AMP will transmit the Minimum Set of Data (MSD) using the modem methods and format specified in these standards. The eCall MSD will (in accordance with Appendix 1) include at least: -

  - Activation: manual (SOS button) or automatic
  - Vehicle Class: passenger Vehicle
  - Vehicle identification number (VIN) – which will be used by EAs to derive Make and Model.
  - Vehicle propulsion storage type (This is important particularly relating to fire risk and electrical power source issues (e.g. Gasoline tank, Diesel tank, Compressed natural gas (CNG))
  - Vehicle location: The last known vehicle's position (latitude and longitude)
  - Confidence in position: set to "Low confidence in position" if the position is not within +/-150m with 95% confidence
  - Direction: bearing

3.1.2 Management of In-Vehicle Systems

Vehicle Manufacturers (or aftermarket providers) are responsible for managing their In-Vehicle Systems (IVS) to prevent spurious or false emergency calls causing problems for the CHA/PSAP and the Emergency Authorities, either directly or through TPSPs with whom they have commercial arrangements

IVS units generating repeated spurious calls or data messages need to be investigated as soon as identified by the CHA/PSAP, so 24-hour reporting must be in place with TPSP/Vehicle Manufacturer/Aftermarket Provider to avoid impacting the handling of genuine emergency calls.
The TPSP/Vehicle Manufacturer/Aftermarket Provider will make checks by calling back the vehicle, the registered keeper (on other numbers), or garage that sold/is selling/or maintains the vehicle: -

- if it is established the unit is faulty then it will be immediately disabled until fixed
- if it is established that the unit is no longer supported by a maintenance agreement then the unit will be immediately disabled until maintenance arrangements are in place.
- if no contact can be made, or the unit cannot be disabled remotely in other ways, then the TPSP/Vehicle Manufacturer/AfterMarket Provider will arrange with the mobile network SIM provider concerned to temporarily suspend or block outgoing service for the SIM card (if this is practical without leading to limited service state emergency calling) until contact can be made (or a restricted service provided to a VM's customer service number only.)

Note

The 999/112 PSAP will need to take temporary measures to manage any influx of spurious calls from an IVS unit, for example by holding the voice call open for as long as possible (to prevent automatic redialling), or by temporarily suspending all data connectivity for the EDSP/TPSP in question if repeated data messages cannot be inhibited to protect service for other callers.

Vehicle Manufacturers (or aftermarket providers) are responsible for managing their In-Vehicle Systems (IVS) over the life of the vehicle to prevent spurious or false emergency calls causing problems for the CHA/PSAPs and the emergency authorities.

3.2 Network Operator Responsibilities

- The mobile NO should deliver the voice call as soon as possible after the SOS-Alert has triggered the call
- Where an SOS Alert voice call needs to be delivered directly to the CHA's network, it needs to be as a 998 call (not 999 or 112) in a manner that ensures the call set-up message carries the caller's telephone number (MSISDN) and the digits 998IIABCD or 998IIEFGHIJKLMN, where II is the network identifier and ABCD is the zone code or EFGHIJKLMN the cell identity.
- Where an eCall needs to be delivered directly to the CHA's network, it needs to be as a 991 call (not 999 or 112) for manually generated calls, or as a 992 call (not 999 or 112) for automatically generated calls, in a manner that ensures the call set-up message carries the caller's telephone number and the digits 991IIABCD or 991IIEFGHIJKLMN, where II is the mobile network identifier and ABCD is the zone code or EFGHIJKLMN the cell identity.
- The NO should deliver a data message according to its agreed performance criteria. For example, a typical time from receipt of data message by NO into the SMSC, to the NO sending the message to the EDSP of less than 10 seconds is desirable.
- The NO will provide a 24 hour data message contact number to any EDSP with which it has specific arrangements for Telematics for use in the event of the EDSP failing to receive data messages. This route will be used to liaise over all operational problems.

3.3 EDSP Responsibilities

- The EDSP will process the data message (for example SMS) into the format agreed by the CHA and the EDSP.
- The data message shall be forwarded to the CHA within an average of 4 seconds from receipt of the data message from the NO using the processes agreed by the CHA and EDSP.
- ·The data message passed on to the CHA from an SOS-Alert call should include :-
    - the Geodetic map reference in the required format,
    - the CLI (telephone number) of the GSM phone
    - an indication of whether the alert was generated manually or by a crash sensor and, using reasonable endeavours, wherever appropriate :-

  - for eCalls, the information within the MSD (see 3.1)
  - for other Telematic SOS calls, the Vehicle's make, model, colour and registration number (or VIN), the registered consumer's/subscriber's name and information on the crash sensor details (airbag, roll over, front, back)

- The EDSP will provide a 24-hour telephone contact to a Support Desk for the CHA in the event of the CHA failing to receive data messages. This route will be used to liaise over all operational problems including town location when data only message received by CHA. The EDSP will use reasonable endeavours to monitor all interface channels and processes to ensure that the data messages are being delivered to the CHA.

- Where data messages fail to reach the CHA, the EDSP should arrange for verbal transfer of the data by contacting the CHA's Lead Centre

- The EDSP will maintain an auditable record of all data messages received from the NO, and those forwarded on to the CHA, for a minimum of 3 months from date of receipt.

3.4 CHA Responsibilities

- The CHA will set-up the voice call within an average of 10 seconds (after answer and after the voice channel is open) to the requested EA, and to the Police if no EA is requested.
- The CHA will also; either match the voice call with the data message from the EDSP when it arrives or be able to decode an MSD sent for an eCall prior to opening the voice channel, and pass the map reference location information to the EA, either using its automated Location Service (EISEC) or through various verbal options: (i) as soon as the EA answers if already received or (ii) by interrupting the EA and consumer when data received (iii) when the EA requests it or (iv) by allowing a Third Party Call Centre operator to pass it where one is being used.
- The CHA will also pass any other details automatically received from the EDSP, or be able to decode an eCall MSD, concerning the location of the incident or the incident itself either automatically via EISEC, or verbally (unless all information required by the EA has been obtained direct from the consumer or a Third party Call Centre operator by the EA).
- The CHA will provide a 24 hour contact number for the EDSP to report any problems with data transmission. This route will be used to liaise over all operational problems.
- The CHA will maintain accessible records of the data messages and the voice messages for 3 months from date of receipt.
- In the event that only a data message is received (with no matching voice call) for a Telematics SOS call, the CHA will search for a voice call. If a voice call has previously been connected, the BT Advisor will connect back to the same Police force. If it is within 30 minutes of a cancelled call, no connection is made. If it is more than 30 minutes, the BT Advisor will attempt to establish voice contact to check if an emergency response is required. If unable to establish voice contact, the BT Advisor will initiate a voice call to the Police and pass all available details.

In the event that an automatically generated eCall is received (with no caller responding after the voice channel is opened), then the BT Advisor will initiate a voice call to the Police and pass all available details.

- The CHA will provide a technical fault reporting telephone number, unique EDSP id and reporting code to allow reporting of technical faults. (If fault relates to failure of data transmission verbal handover of details from the EDSP should separately take place (see 3.3) in parallel with reporting of the fault).

4 Additional Requirements for SOS-Alert Services, or eCall Services, using a Third Party Call Centre/Third Party Service Provider [introduced in 2013].

During 2013 further discussions between the 999 Liaison Committee, EAs and the CHA took place, following on from requests to support a telematics service where the voice call would connect directly to a nominated third-party call centre instead of direct to the CHA. Further consideration led to the following requirements: -

A third-party telematics service provider (TPSP) is expected to provide a call answering service at a 24-hour Call Centre as well as to fulfil the EDSP role.

The nominated third-party call centre (TPCC) provides a specific telephone number for receipt of calls directly from the vehicle. Appropriate resilience is expected to be provided through use of more than one TPCC and/or by the in-car IVS unit falling back to make a standard 112/999 call if no answer from the TPCC. The TPCC is normally expected to be UK-based but can be in other European countries.

The TPCC is responsible for quickly answering calls and then determining if an emergency response is required – where doubt exists calls will be extended through the CHA to the Police.

According to eCall standards (EN16102:2011 mentioned in Annex A), answering 90% of calls within 15 seconds provides an adequate level of quality. The normal standard for the UK's CHA is to answer 95% of the calls within 5 seconds

The TPCC will connect the call to the CHA via a specific number with minimal delay and its operators will follow an agreed procedure to allow the CHA to set up a voice contact with the relevant EA.

The TPCC will allow direct voice conversation between the vehicle occupants and the EA when requested.

The TPCC will set-up and use data links to the CHA that allow incident and location data to be automatically transferred to the EA as soon as the voice call is answered - as detailed in Section 3.3 (EDSP Responsibilities) above. As a temporary measure (eg during service establishment when volumes very low) data can be transferred verbally by the TPCC

The TPCC will provide additional information and assistance to the CHA and EA, including language translation if the caller cannot speak English, to help the EA organise an appropriate response.

The TPCC will ensure that when a customer signs up to a TPCC telematics service, they acknowledge that their information can be used and passed to the emergency services for the delivery of the service.

4.1 Voice only services

During 2020, BT started to receive requests from TPCCs advising of discussions with Motor Manufacturers to handle their vehicles' in-car device activations. Where any of these activations needed an emergency response, the TPCC would need to connect to BT for onward connection to the EAs. These TPCCs wanted to be able to pass all information verbally, due to anticipated low volumes and difficulties associated with electronic transfer of data.

Following on from previous agreement that TPCC service providers would be allowed to temporarily transfer data information verbally during service establishment, it has been recognised that call volumes have continued to remain at a low level. Also, there have not been call handling issues arising when dealing with verbally passed data. In addition to continued low volumes, many TPCCs are based outside the UK, which presents technical challenges with regards to transferring the MSD associated with an eCall.

At the 999 Liaison Committee held on 30 June 2021, the introduction of support for a Voice Only Service from a TPCC was discussed and agreed to. It was suggested that BT should monitor and report back to the 999LC should any call handling issues arise, particularly if call volumes become high, so continued support of voice only services could be revisited

5 Liability

Each of the parties involved in connecting the GSM-enabled, location-based device to the EAs will make reasonable endeavours to convey all agreed information relevant to an SOS-Alert call/eCall, whether by voice or by data message, in a timely and accurate manner, and in accordance with the relevant UK and European Data Protection and Telecommunication Acts.

6 General

Any changes to this MoU or related documents, including software upgrades that impact the performance of MoU roles, would need 3 months' notice of the intention to change followed by agreement between the affected parties on the correct implementation plan. All changes are to be backward compatible.

7 Change Control History

Draft 1: issued for comment April 1999 to AA, RTT, BT Cellnet and Vodafone

Draft 2: issued 7 June after discussion with AA, BT Cellnet and Vodafone

Draft 3: issued 16 June 1999 after further discussions with AA, BT Cellnet and Vodafone. Circulated to CWC, RTT, Mannesmann and Tegaron for comment.

Issue 1: published on 9 July after further meeting with AA, BT Cellnet and Vodafone.
Issue 2: published on March 31 2000 following comments from industry

Issue 3: published on 10th April 2000 following further comments from industry

Issue 4: published on 18th May 2000 following further comments from industry

Issue 5: published on 25th May 2007

Issue 6: published November 2008

Issue 7: published March 2009

Issue 8: published Jan 2011 (remove ref 15999)

Issue 9: published Jan 2012: update EDSPs, add fault reporting, add accidental call precaution and clarify text at various points.

Issue 10: draft issued for comment July 2014

Issue 10: published as an Annex in PECS CoP in October 2014

Issue 11: draft with eCall proposals and voice only services issued for comment June 2021

Issue 11: published July 2021

8 Glossary of Terms

VASS - Value Added Service Supplier
NO - GSM Network Operator.
EDSP - Emergency Data Service Provider
CHA - SOS Call Handling Agency/ Designated eCall PSAP
EA - Emergency Authorities: the Fire, Police, Ambulance and Coastguard services
GSM - Global Standard for Mobile phones
GPS - Global Positioning by Satellite
TPCC - Third Party Call Centre
TPSP – Third Party Service Provider
MSD – Minimum Set of Dat

Annex 1 eCall standards

EN 15722:2015 Intelligent transport systems - ESafety - ECall minimum set of data

EN 16062:2015 Intelligent transport systems - ESafety - eCall high level application requirements (HLAP) using GSM/UMTS circuit switched networks
EN 16072:2015 Intelligent transport systems - ESafety - Pan-European eCall operating requirements

EN 16102:2011 Intelligent transport systems - eCall - Operating requirements for third party support

This publication is available from: telecoms.resilience@dsit.gov.uk

If you need a version of this document in a more accessible format, please email alt.formats@dsit.gov.uk. Please tell us what format you need. It will help us if you say what assistive technology you use.